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Redox evolution of crystallizing magmas with C-H-O-S volatiles and its implications for atmospheric oxygenation

Chenguang Sun [a,*], Cin-Ty A. Lee [b]

[a] Department of Geological Sciences, Jackson School of Geosciences, University of Texas at Austin, Austin, TX 78712, USA
[b] Department of Earth, Environmental and Planetary Sciences, Rice University, Houston, TX 77005, USA

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ABSTRACT

The evolution of Earth's atmospheric O_2 levels on long timescales is modulated in part by the composition of magmatic gases, specifically their oxidation state and speciation. Here, we examine how magma differentiation affects the oxygen fugacity and composition of magmatic gases. Variations in temperature, pressure, composition, and redox conditions of evolving magmatic systems result in complexities, which have not been fully treated previously. Taking all these variables into account, we develop a multipronged approach to a comprehensive understanding on the role of magmatic gases on atmospheric oxygenation. This includes a compilation of petrological observations on the redox systematics of mid-ocean ridge basalts, a geochemical model for magma crystallization in oxygen-closed systems with variable initial redox conditions, and a thermodynamic framework for gas–melt systems with multi-component C-H-O-S volatiles.

By considering magmatic differentiation at variable depths, we show that magma crystallization progressively changes the redox state of residual melts and coexisting gases due to differences in iron speciation between the melt and cumulate minerals. In thick crusts, high pressures of magma differentiation suppress magnetite saturation but promote garnet crystallization, increasing magmatic oxygen fugacity by ~4–8 orders of magnitude and thereby resulting in oxidized gases. In thin crusts, the prevalence of magnetite prevents significant oxidation of the magma, yielding reduced gas emission. Hence, magmatic gases from thick crusts are much less effective O_2 sinks, which does not appear to be affected by gas decompressional transport or polybaric degassing from the deep crustal magmatic zone to the atmosphere. Depending on how crustal thickness during active magmatism varies with time, magmatic gases may play an important role in modulating atmospheric O_2 levels on million-year timescales. Rapid growth of continental crust in the late Archean appears to have been associated with the first evidence for significant crustal thickening during magmatism. If so, this would have led to a substantial decrease in O_2 sink, possibly permitting atmospheric O_2 to rise to a new state.

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1. Introduction

Through most of Earth's history, atmospheric O_2 has been generated by oxygenic photosynthesis. The rise of atmospheric O_2 to an appreciable level occurred ~2.5–2.0 Ga, a transition known as the Great Oxidation Event (GOE). Yet, this time interval was long after the emergence of O_2-producing organisms at ~3 Ga (e.g., Farquhar and Wing, 2003; Canfield, 2005; Lyons et al., 2014; Planavsky et al., 2014). As noted in previous studies (e.g., Holland, 2002, 2009; Kump and Barley, 2007; Gaillard et al., 2011; Kasting, 2013; Lee et al., 2016; Moussallam et al., 2019; Eguchi et al., 2020), secular changes in the rates of O_2 production or variations in the nature of O_2 sink are required to explain the rise of atmospheric O_2. Magmatic degassing, in particular, has long been regarded as an important O_2 sink (e.g., Kasting et al., 1993; Holland, 2002). Because magmatic gases typically include reduced C, H, and S species (e.g., CO, CH_4, H_2, H_2S, and S_2), they could rapidly react with O_2 upon release to the atmosphere. Thus, a popular view is that secular decrease in the global flux of reduced magmatic gases may have played an important role in delaying the rise of atmospheric O_2.

Several hypotheses have been proposed to explain the possible causes for the decrease in the global flux of reduced magmatic gases. These include (1) a secular increase in mantle oxygen fugacity (fO_2) (e.g., Kasting et al., 1993; Aulbach and Stagno, 2016;

* Corresponding author.
E-mail address: csun@jsg.utexas.edu (C. Sun).

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Nicklas et al., 2019; Kadoya et al., 2020a), (2) a decrease of volcanic emission depths from submarine to subaerial (Kump and Barley, 2007; Gaillard et al., 2011), or (3) cooling of volcanic gases in closed gaseous systems (Moussallam et al., 2019). Hypothesis (1) contrasts with earlier studies (e.g., Canil, 1997; Delano, 2001; Lee et al., 2003; Li and Lee, 2004) that suggest a near constant upper mantle fO_2 since the early Archean, but is supported by recent constraints on mantle fO_2 (Aulbach and Stagno, 2016; Nicklas et al., 2019). According to these recent studies, the apparent increase in mantle fO_2 from the Archean to Proterozoic was about one log unit. However, a fundamental assumption in Hypothesis (1) is that the magmatic gas fO_2 is buffered by Earth's upper mantle, which is unlikely to be the case after magmas leave their mantle source regions. Hypothesis (2) focuses on volcanic degassing during shallow eruption (~1–100 bar) without changes in temperature (T) and melt compositions (except for Fe^{2+}/Fe^{3+}). This process could significantly increase SO_2/H_2S ratios but, as discussed by Kasting (2013), may not adequately change the total redox budget of the gases to affect atmosphere O_2. Hypothesis (3) emphasizes the oxidation of volcanic gases during cooling in closed systems. However, it has also been suggested that the cooling gases may become more reduced if buffered by melts (e.g., Burgisser et al., 2008; Kadoya et al., 2020b). Importantly, all these hypotheses are likely oversimplified because they focused on either gas-only systems or gas–melt systems with fixed melt compositions and, in many cases, buffered fO_2.

Natural magmatic systems are more complicated, because they evolve dynamically and chemically with multiple phases, including crystals, melt, and gas (Fig. 1). Volatile elements in the magmatic systems are initially extracted from Earth's deep interior as dissolved species in silicate melts. After emplaced in the deep crustal magmatic zone, mantle-derived melts undergo cooling and decompression, leading to crystallization and degassing. Crystallization could systematically change the residual melt composition and enrich the volatile elements in the melt. The total volatile budgets in the magmatic system, however, can be reduced by saturation of graphite, sulfide, and/or sulfate in addition to volatile degassing. During degassing, the residual melt establishes redox and chemical equilibria with the coexisting gas. The extent of gas–melt interactions is determined by volatile solubilities in silicate melts, which often depend on T, pressure (P), melt composition, and particularly fO_2. If fO_2 of the system is not buffered, fractionation of redox-sensitive elements (e.g., Fe, S, C, and H) by crystallization or degassing could also yield redox changes and in turn affect the gas composition. Such complexities have long been noted in the literature (e.g., Byers et al., 1984; Holloway, 1987; Carmichael, 1991; Holloway and Blank, 1994; Moretti and Papale, 2004; Burgisser and Scaillet, 2007; Mangiacapra et al., 2008; Oppenheimer et al., 2011; Iacovino, 2015; Cicconi et al., 2020). However, an integrated view of dynamically evolving magmatic systems with multi-component C-H-O-S volatiles during solidification and degassing has not been established. Thus, it remains largely unknown how differentiation of mantle-derived melts affects gas composition.

Here we adopt a multipronged approach to explore how gas composition changes during magma crystallization and degassing at variable depths. First, we use petrological and geochemical observations to quantify the magnitude of fO_2 variations during magma crystallization. These observations are combined with petrological modeling to gain insight into the nature of crystallizing assemblages and the systematics of fO_2 variations. We then build a comprehensive thermodynamic framework to investigate the chemical and redox interactions between solidifying magmas and multi-component C-H-O-S gases. This framework allows us to model the melt fO_2 and gas composition during magma crystallization in the deep crust and during polybaric degassing from the deep crust to surface. We show that during magma crystallization, the fO_2 of residual melts and coexisting gases is primarily determined by differential fractionation of oxidized- (Fe_2O_3) and reduced-iron (FeO) species. Decompression transport of closed gases or polybaric degassing of evolved magmas could further modify the fO_2 but does not have a significant effect on the overall reducing power of the bulk gas from the evolved magma. Noting that the mineralogy of crystallizing assemblages is sensitive to the average depth of crystallization, we then demonstrate that

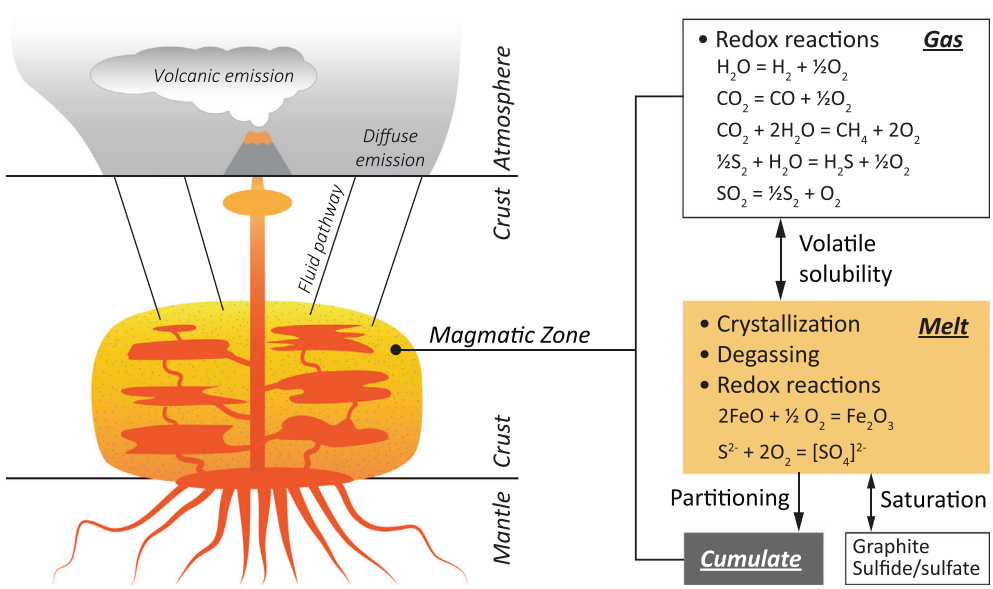

Fig. 1. Schematic illustration of a dynamically evolving magmatic system with C-H-O-S volatiles. In the deep crustal magmatic zone, mantle-derived melts are subject to fractional crystallization and volatile degassing and subsequently develop into three phases including a residual melt, a multi-component gas, and cumulate minerals. Segregation of cumulate minerals form the igneous crust while changing the residual melt composition. Upon release to the surface, the magmatic gas can immediately affect the atmosphere O_2 level by consuming O_2 through reduced volatile species (e.g., H_2, CH_4, CO, H_2S, and S_2). During most of the crystallization process, the gas and residual melt maintain equilibrium through redox reactions of volatile species in the gas, dissolution of volatiles in the melt, and redox reactions of Fe^{2+}/Fe^{3+} and S^{2-}/S^{6+} in the melt. Saturation of graphite, sulfide and/or sulfate could also modify the volatile element inventory in the gas–melt system. These redox and chemical reactions determine the abundances of reduced volatile species and thus control the efficiency of O_2 removal by magmatic gases.

the nature of crystallization places a first order control on magma fO_2, the composition and reducing power of magmatic gases, and ultimately Earth's atmosphere oxygenation.

2. Observations of magma redox evolution

2.1. Data compilation

The oxidized- to total Fe ratio ($Fe^{3+}/\Sigma Fe$) in silicate melts is known as a reliable proxy for fO_2 (e.g., Kress and Carmichael, 1991; Cottrell and Kelley, 2011; O'Neill et al., 2018). To explore how melt fO_2 evolves during crystallization, we compiled 413 mid-ocean ridge basalt (MORB) glasses with Fe speciation information from the literature (Byers et al., 1983, 1984; Christie et al., 1986; Wallace and Carmichael, 1992; Bézos and Humler, 2005; Berry et al., 2018; Birner et al., 2018; Zhang et al., 2018). We only considered MORB samples in that they could better represent the fractionation trend and are less affected by significant source heterogeneities and/or lithosphere contamination than ocean-island basalts or arc lavas. Fe speciation of these samples was analyzed by wet chemistry methods (N = 237) or in situ X-ray absorption near edge structure (XANES) spectroscopy (N = 176). The compilation (Table S1) covers a broad range of compositions (MgO = 1.5 – 12.2 wt%; ΣFeO = 7.2 – 18.1 wt%) with Mg# (=100 × Mg/(Mg + ΣFe) in mol%) of 17.0–73.6.

It has been known that different analytical methods cause systematic discrepancies in the measured $Fe^{3+}/\Sigma Fe$ ratios of MORBs (e.g., Berry et al., 2018; Zhang et al., 2018; Losq et al., 2019; Asimow, 2021; Bézos et al., 2021). Because MORB glasses often contain olivine microcrystals, it has been suggested that wet chemical analyses likely underestimate the $Fe^{3+}/\Sigma Fe$ ratio (e.g., Cottrell and Kelley, 2011). In situ XANES studies are thought to avoid the olivine microcrystals but there are differences in reported $Fe^{3+}/\Sigma Fe$ ratios between different research groups as a result of different calibration approaches (e.g., Berry et al., 2018; Zhang et al., 2018). Differences between wet chemical and XANES studies are small, but significant enough to lead to different interpretations of mantle melting processes (e.g., Asimow, 2021). The primary goal of our study is not to resolve these differences and mantle redox, but to characterize the overall variation of melt fO_2 during magmatic differentiation. More evolved MORBs (e.g., MgO < ~4 wt%) are particularly important for our purpose but were only reported in earlier wet chemistry studies (Byers et al., 1983, 1984; Christie et al., 1986; Wallace and Carmichael, 1992).

Given the inter-laboratory discrepancies in measured $Fe^{3+}/\Sigma Fe$ ratios, we adjusted all $Fe^{3+}/\Sigma Fe$ data to be consistent with the most recent XANES calibration of Berry et al. (2018) that is also consistent with the wet chemistry studies (e.g., Bézos et al., 2021). Details of the correction schemes are outlined in Appendix A.1. We stress that such corrections are not intended to justify any calibration or analytical methods but instead to highlight the overall variations of $Fe^{3+}/\Sigma Fe$ ratios during MORB differentiation. The model of Kress and Carmichael (1991) was then used to calculate the fO_2 for the compiled samples (see details in Appendix A.1). Liquidus T's at 2 kbar of these samples were obtained by applying the magma thermometer of Sun and Dasgupta (2020). The calculated fO_2 was compared with that buffered by the quartz-fayalite-magnetite (QFM) assemblage on a log scale. The difference was defined as $\Delta QFM = \log fO_2 - \log fO_2[QFM]$. For consistency with our crystallization modeling in Section 3, we used the QFM buffer from Myers and Eugster (1983). Detailed results of the $Fe^{3+}/\Sigma Fe$ corrections, ΔQFM, and liquidus T are listed in Table S1 and shown in Fig. 2. The calculated ΔQFM values at 2 kbar range from −2.0 to +1.3, except for four primitive samples (MgO = 8.1 – 9.7 wt%; ΔQFM = −3.1 to −2.6). These four samples are from wet chemistry studies, including 3 from Bézos and Humler (2005) and 1 from Byers et al. (1984). Given their significantly lower fO_2, these four samples were considered as outliers and hence excluded from Fig. 2a-b.

2.2. Redox systematics of MORB melts and key questions

The compiled MORB glasses display large variations in $Fe^{3+}/\Sigma Fe$ ratios (0.08 – 0.26; Fig. 2a) corresponding to about two orders of magnitude variation in fO_2 (ΔQFM = −1.5 to +1.8 at 2 kbar; Fig. 2b). Although the most primitive samples (MgO ≥ 8 wt%) show some extents of heterogeneity in mantle fO_2, $Fe^{3+}/\Sigma Fe$ ratios and $\log fO_2$ vary with liquidus T and show abrupt kinks at ~1100 °C (i.e., MgO = ~4 wt%). This suggests an effect of magma differentiation on the redox systematics of MORB melts. During early differentiation ($T \geq$ ~1100 °C or MgO ≥ ~4 wt%), $Fe^{3+}/\Sigma Fe$ ratios of MORB glasses increase by up to a factor of ~2–3, corresponding to the increase of melt fO_2 by ~2 orders of magnitude. With further differentiation, the $Fe^{3+}/\Sigma Fe$ ratio and $\log fO_2$ of the melt gradually decrease towards those of the primitive MORBs. This demonstrates that MORB melts undergo auto-oxidation during early-stage differentiation followed by auto-reduction during late-stage differentiation. Such a redox transition is inconsistent with a fixed fO_2 buffer during magma differentiation. Notably, the extent of fO_2 variation is about one order of magnitude greater than any proposed secular increase of upper mantle fO_2 since the Archean (Aulbach and Stagno, 2016; Nicklas et al., 2019). As fO_2 plays an important role in vapor equilibria and gas–melt interactions, the redox effect of magma differentiation must be considered to assess the influence of magmatic gases on the atmospheric oxygenation.

The primary cause of redox variations during magma differentiation is a fundamental problem in petrology (e.g., Mathez, 1984; Carmichael, 1991). It has been suggested that crystallization of Fe^{3+}-poor mafic silicate minerals is responsible for the increase of Fe^{3+} in the melt (e.g., Carmichael, 1991; Lee et al., 2010; Kelley and Cottrell, 2012; Tang et al., 2018). Because Fe is a major element in silicate melts with two different valence states (Fe^{2+} and Fe^{3+}), the increase of Fe^{3+}/Fe^{2+} ratios by crystallization can lead to auto-oxidation if the magmatic system is closed to oxygen exchange with the surrounding. Degassing of C- or H-species has also been argued as a major process to significantly oxidize silicate melts during shallow eruption (e.g., Mathez, 1984; Burgisser and Scaillet, 2007). On the other hand, melt reduction, as indicated by the decrease of Fe_2O_3 in silicate melts, has been attributed to either fractionation of Fe oxide minerals (e.g., Byers et al., 1983; Carmichael, 1991; Jenner et al., 2012) or degassing of S (e.g., Carmichael, 1991; Burgisser and Scaillet, 2007; Kelley and Cottrell, 2012; Moussallam et al., 2014; Brounce et al., 2017). Because mineral fractionation and volatile degassing are not mutually exclusive (Fig. 1), it is possible that they could both impact the magma redox systematics.

To delineate the primary cause of the redox variation, ΣFeO, S, Fe_2O_3, H_2O and CO_2 in MORB glasses are also plotted in Fig. 2 in comparison with the $Fe^{3+}/\Sigma Fe$ ratio and $\log fO_2$. We found that the $Fe^{3+}/\Sigma Fe$ ratio and $\log fO_2$ mimic the trends of ΣFeO, S, and Fe_2O_3 and display abrupt kinks at ~1100 °C (MgO = ~4 wt%). H_2O shows progressive enrichment by over one order of magnitude (Fig. 2f), consistent with fractional crystallization and small extents of degassing. CO_2 appears to be independent of crystallization (cf. circles in Fig. 2f) due to extensive degassing. Given that the differentiation trends of H_2O and CO_2 are distinct from that of Fe_2O_3, their degassing likely has no major impact on melt fO_2. The positive correlation between S and Fe_2O_3 in basaltic melts has been used previously as evidence for the effect of SO_2 degassing on melt reduction (e.g., Kelley and Cottrell, 2012; Moussallam et al., 2016).

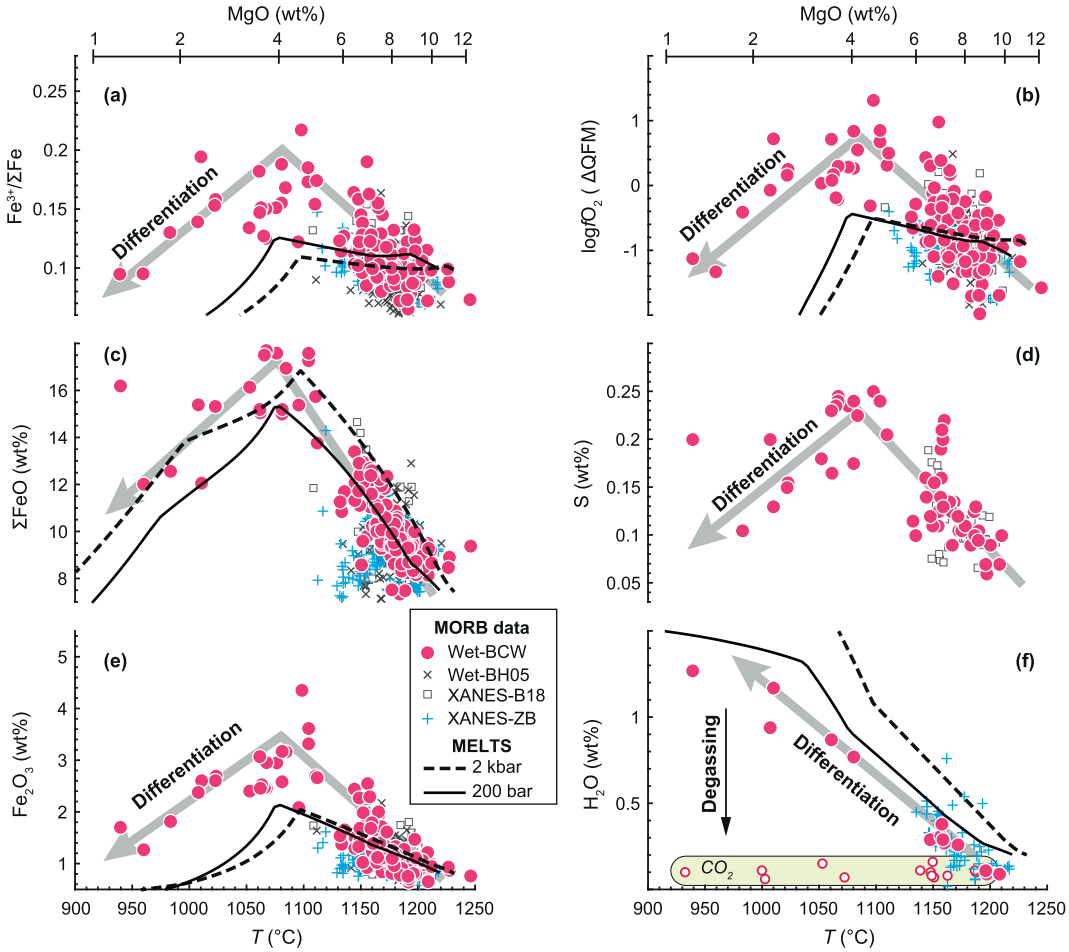

Fig. 2. Redox (a-b) and composition (c-f) variations of MORB glasses as a function of liquidus temperature (*T*). MgO on the top axis corresponds to the liquidus *T* on the bottom. MORB differentiation (grey arrows) results in the kink of $Fe^{3+}/\Sigma fe$ ratios (a) that determines the redox transition from auto-oxidation to auto-reduction as manifested by the variation of $\log fO_2$ (b). This redox transition is coupled with the variations of total FeO (c), S (d), and Fe_2O_3 (c) but is decoupled with the trends of H_2O and CO_2 dissolved in the glasses (f). "Wet-BCW" denotes the wet chemical data from Byers et al. (1983, 1984), Christie et al. (1986), and Wallace and Carmichael (1992). "Wet-BH05" indicates the wet chemical data from Bezos and Humler (2005). "XANES-B18" shows the XANES data from Berry et al. (2018). "XANES-ZB" denotes the XANES data from Zhang et al. (2018) and Birner et al. (2018). High-*T* samples appear to be more abundant because many previous studies focused on the primitive MORB glasses to investigate mantle fO_2. Solid and dashed curves show the results from MELTS crystallization modeling without a fixed fO_2 buffer at 200 bar and 2 kbar, respectively. The initial melt composition for MELTS modeling is listed in Section 3. Note that the MELTS modeling reproduces the fractionation of major elements in MORBs (Fig. 2c) but cannot generate the redox variation (Fig. 2a-b) of MORBs due to its significant overestimation of Fe_2O_3 partitioning in cumulate minerals (Fig. 2e).

However, fractionation of sulfide, which is often observed as an immiscible blobs in MORBs and cumulate rocks (e.g., Francis, 1990; Patten et al., 2012), may also play a role. The coupled decreases of S and FeO in the evolved MORBs (Fig. 2) are consistent with the effect of FeO on S^{2-} contents at sulfide saturation (SCSS; e.g., Ding and Dasgupta, 2017; Smythe et al., 2017). Because FeO depletion requires the fractionation of Fe-rich minerals, the co-variations among FeO, Fe_2O_3 and S in MORB glasses suggest that the change of cumulate mineral assemblages likely controls the redox systematics and S variations in the evolving melts. With Fe_2O_3 and FeO both as major components, magnetite seems to be a critical mineral responsible for the observed redox transition in MORB melts (e.g., Byers et al., 1983; Carmichael, 1991; Jenner et al., 2012). We emphasize that these effects of fractionation on fO_2 as magmas differentiate do not negate the possibility of spatial variations in mantle fO_2 (e.g., Cottrell et al., 2021; and references therein), but are superimposed on these source variations.

Although our compiled MORB samples indicate a major control of crystallizing mineral assemblages on fO_2, several questions remain to be answered for a comprehensive understanding of the redox systematics and gas compositions in evolving magmatic systems. It is still unclear how cumulate fractionation can affect the melt fO_2 under conditions distinct from MORBs; to what extent volatile degassing can alter melt fO_2; whether the variation of mantle fO_2 can override the redox effect of magma differentiation; and ultimately how gas composition varies with magma differentiation for variable crystallization conditions and mantle fO_2. To address these questions, we build a new quantitative approach in the following two sections. This includes (1) a geochemical model in Section 3 for solidifying magmatic systems closed to oxygen and (2) a thermodynamic framework in Section 4 for evolving magmatic systems with multi-component C-H-O-S volatile species. The first model allows us to simulate melt redox systematics during fractional crystallization at a broad range of initial fO_2 and depths, while the second model enables us to assess the gas–melt interactions and the change of gas compositions during crystallization and degassing.

3. Crystallization-driven redox evolution

3.1. Model setup

In this section, we seek to model the redox effect of magma crystallization. For simplicity, second order processes (e.g., magma mixing, recharge, and crustal assimilation) were not considered.

The magma chamber was approximated as an oxygen-closed system. The fO_2 of the magmatic system is controlled by the relative abundances of FeO and Fe_2O_3 in the melt. In general, FeO and other major oxide compositions of the melt can be modeled using existing thermodynamic programs, among which the MELTS program (Ghiorso et al., 2002; Smith and Asimow, 2005) has been widely used. These programs, however, have large uncertainties in modeling the partitioning of Fe_2O_3 (e.g., Davis and Cottrell, 2018; Rudra et al., 2021; Rudra and Hirschmann, 2022) and thus have difficulty in simulating redox change during magma crystallization in systems not buffered in fO_2. As demonstrated in Fig. 2, results from MELTS calculations can reproduce the fractional trends of MORB composition (e.g., ΣFeO and H_2O) by assuming low-*P* fractional crystallization at 0.2 and 2 kbar without a fixed fO_2 buffer, which ensures the accuracy of the MELTS program to predict phase equilibria and major oxide compositions of the melt. The MELTS results also confirm the redox transition of MORBs at ~4 wt% MgO, but the modeled variations of $Fe^{3+}/\Sigma Fe$ ratios and $\log fO_2$ are much smaller than those of compiled MORBs by at least a factor of two to three. This is because the MELTS program systematically overestimates the partitioning of Fe_2O_3 in minerals, as demonstrated experimentally by Davis and Cottrell (2018) and Rudra and Hirschmann (2022), and hence suppresses the increase of Fe_2O_3 and $Fe^{3+}/\Sigma Fe$ ratios in the residual melt.

To improve the MELTS simulation for melt Fe_2O_3 and redox variations, we modified the MELTS calculation by incorporating more accurate partition coefficients (*D*) of Fe_2O_3. We corrected the $Fe^{3+}/\Sigma Fe$ ratio of the residual melt at each step of crystallization and then used the corrected melt composition and fO_2 as inputs for MELTS to calculate the next step of crystallization. This allows us to make use of the rigorous phase equilibria calculations in MELTS, as it can be well tested by the major element compositions of MORBs (Fig. 2). We note that the *D* values of Fe_2O_3 (= 0.06 – 0.4) for silicate minerals (e.g., Mallmann and O'Neill, 2009) are several orders of magnitude smaller than that for magnetite (*D* = 5 – 159; Table S2). Given the large variation of Fe_2O_3 compatibility in magnetite, an accurate model for Fe_2O_3 partitioning between magnetite and silicate melt is essential for modeling the fractionation of Fe_2O_3 during crystallization. As detailed in Appendix A.2, we developed a simple empirical model that could well reproduce the experimentally measured *D* values for Fe_2O_3 in magnetite. Given the incompatibility of Fe_2O_3 in silicate minerals, we assumed a constant bulk *D* value (= 0.1) for the cumulate assemblage excluding magnetite.

Combining the new partitioning parameters with the MELTS program, we modeled the fractionation of Fe_2O_3 and FeO during fractional crystallization through an incremental stepwise approach. This is an iterative procedure with two basic steps: (1) For an initial fO_2, we used the MELTS program to determine the *T*, phase relations, and melt composition at the first increment of crystallization; (2) using results from Step (1), we then calculated the bulk *D*'s of Fe_2O_3 in the cumulate assemblage and computed the $Fe^{3+}/\Sigma Fe$ ratio of the residual melt. The MELTS calculation considered a minimum *T* of 800 °C. The fO_2 was then determined by applying the model of Kress and Carmichael (1991). Results from Step (2) were taken as new inputs for Step (1) to compute the next increment of crystallization. To partially automate this procedure, we run the MELTS program for a wide range of buffered fO_2 (ΔQFM = −3 to +8) and compiled the results in a multi-dimensional grid. The two-step iterative calculations were then conducted by interpolating *T*, melt compositions, and phase proportions with adaptive changes of fO_2 for every 1 % of crystallization.

The initial melt in these calculations took the composition of a primitive MORB glass (AII107-D20-1) from Zhang et al. (2018), which contains 49.72 wt% SiO_2, 0.63 wt% TiO_2, 17.52 wt% Al_2O_3, 7.47 wt% ΣFeO, 9.96 wt% MgO, 13.01 wt% CaO, 1.65 wt% Na_2O, 0.02 wt% K_2O, and 0.12 wt% P_2O_5. Prior to crystallization, the initial melt was set at four different redox conditions (ΔQFM = −2, −1, −0.5, 0, and 1) corresponding to mantle fO_2 variation. Four pressures (0.2, 2, 9, and 15 kbar) were considered to cover a broad range of crystallization depths. As indicated by MORBs and arc lavas (e.g., Arevalo and McDonough, 2010; Plank and Manning, 2019), the initial melts from thin and thick crusts likely contain variable amounts of dissolved H_2O. Taking these as analogs, we assumed 0.2 and 4 wt% H_2O in the initial melt for crystallization at low- (0.2 and 2 kbar) and high-*P* (9 and 15 kbar), respectively. Results of our crystallization modeling are detailed in Table S3.

3.2. Modeling results

We observed systematic variations of cumulate mineral assemblages during crystallization at variable depths and initial fO_2 (Fig. 3). At low pressure (0.2 and 2 kbar), typical cumulate minerals include olivine, pyroxene, plagioclase, and magnetite. Magnetite joins the low-*P* cumulate at ~80 % solidification (melt MgO = ~4–5 wt%) followed by a gradual decrease, while more reduced initial melts appear to suppress magnetite saturation in more evolved melts. At 9 kbar, magnetite becomes saturated at ~30 % crystallization (melt MgO = ~5 wt%), garnet begins to crystallize at ~35 % crystallization (melt MgO = ~4 wt%), whereas plagioclase joins the cumulate at ~60 % crystallization (melt MgO = ~1 wt%). During crystallization at 15 kbar, magnetite and plagioclase do not reach saturation, while garnet starts to crystallize earlier (~25 % crystallization) than that at 9 kbar. Unlike the low-*P* cases, cumulate mineral modes at 9 and 15 kbar are more sensitive to initial fO_2. Magnetite modes at 9 kbar are generally <20 wt% but become greater than 30 wt% for the most oxidized initial melt (QFM + 1). In addition, a more oxidized initial melt at 9 and 15 kbar seems to produce more pyroxene in the cumulate. We note that in the results of our crystallization modeling, amphibole, a common hydrous mineral in arc cumulates, is not a liquidus phase of the high-*P* (hydrous) melt. This is because the MELTS program currently does not have a comprehensive model for amphibole. As we discussed in Appendix A.3, amphibole crystallization unlikely affects the first-order redox variation during high-*P* crystallization.

Melt compositions also show considerable variations during crystallization (Fig. 4a, d, g and j). Specifically, ΣFeO and Fe_2O_3 show distinct transitions from an early enrichment to a subsequent depletion, the magnitude of which strongly depends on *P*. During low-*P* crystallization (0.2 and 2 kbar), the enrichment of ΣFeO in the melt is primarily controlled by the fractionation of plagioclase (i.e., an Fe-poor mineral), while that of Fe_2O_3 can be attributed to its incompatibility in the cumulate. Upon magnetite saturation at low *P*, both ΣFeO and Fe_2O_3 in the melt start to decrease significantly. The depletion of ΣFeO is further enhanced in the highly evolved melt, because FeO becomes enriched in the silicate cumulates (e.g., olivine and pyroxene). Due to the absence of plagioclase (i.e., the Fe-poor mineral) in the high-*P* cumulate, ΣFeO in the melt becomes slightly increased during early-stage high-*P* crystallization. However, the high-*P* melts show a rapid decrease of ΣFeO at ~4–6.5 wt% MgO (i.e., ~30–40 % crystallization), which can be attributed to the crystallization of garnet (i.e., an Fe-rich silicate mineral) and progressive enrichment of FeO in other silicate minerals. We note that the differentiation trends of melt ΣFeO at low- and high-*P* from our model calculations mimic the classic liquid lines of descent for tholeiitic and calc-alkaline series, respectively. Fe_2O_3 in high-*P* melts continuously increases during early-stage crystallization but begins to decrease with the intensified depletion of ΣFeO in the melt. This is because Fe_2O_3 in an oxygen-closed magmatic system is also constrained by ΣFeO in the melt

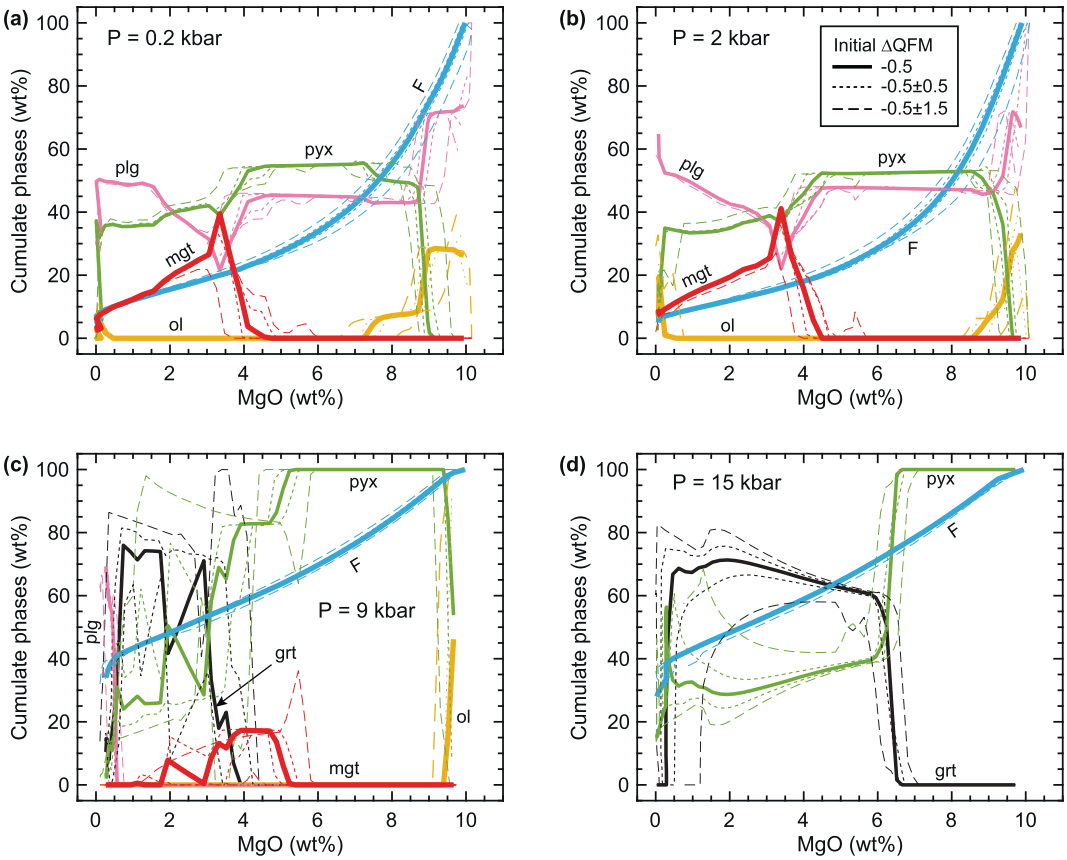

Fig. 3. Modeling results of mineral abundances in the cumulate as a function of MgO in the melt during crystallization at 0.2 kbar (a), 2 kbar (b), 9 kbar (c) and 15 kbar (d). Major minerals in the cumulate include olivine (ol), pyroxene (pyx), plagioclase (plg), magnetite (mgt), and garnet (grt). The residual melt (*F*) is also shown as an index for the extent of crystallization. The initial ΔQFM indicates the redox condition of the initial melt with respect to the QFM buffer on a log scale.

through redox equilibrium reaction, $2FeO + ½O_2 = Fe_2O_3$. The decrease of ΣFeO in the melt can displace this redox reaction towards the left with decreasing Fe_2O_3 but increasing fO_2. Apparently, the rapid decrease of ΣFeO in highly evolved melts (MgO < 1 wt%) at high *P* overturns the effect of silicate mineral crystallization, yielding the decrease of Fe_2O_3 in the melt.

As driven by crystallization, the differential fractionation between ΣFeO and Fe_2O_3 results in systematic variations of melt fO_2 at different depths. During low-*P* crystallization, melt redox evolution appears to have three stages, including (1) an early-stage auto-oxidation, (2) a subsequent auto-reduction, and (3) a final-stage auto-oxidation (Fig. 4b-c and e-f). The early auto-oxidation is simply due to fractionation of silicate minerals, in which Fe_2O_3 are highly incompatible. Given that the first redox transition occurs at magnetite saturation in moderately evolved melts (MgO = ~4 – 5 wt%), it is evident that magnetite crystallization is responsible for auto-reduction in the second stage. Because silicate minerals, such as olivine and pyroxene, become progressively enriched in FeO as they crystallize from more evolved melts (e.g., Martel et al., 1999; Scaillet and Evans, 1999; Sisson et al., 2005), the highly evolved melt (MgO < 1 wt%) show a rapid decrease in ΣFeO (Fig. 4a-b) that effectively increases the Fe^{3+}/ΣFe ratios and fO_2 during late-stage crystallization (i.e., the final-stage auto-oxidation). Such an auto-oxidation is expected to extend to T < 800 °C if the lower T were considered in the crystallization modeling. However, high-*P* crystallization causes continuous auto-oxidation (Fig. 4h-i and k-l) primarily due to the lack of significant magnetite segregation. The short interval of magnetite saturation at 9 kbar (Fig. 3c) slightly suppress the auto-oxidation at ~3–5 wt% MgO (Fig. 4i), but this effect appears to be negligible due to the small abundance of magnetite in the crystallizing assemblage. Compared with low-*P* cumulates, the crystallizing mineral assemblages at high *P* lack plagioclase (i.e., Fe-poor) but contain abundant garnet (i.e., Fe-rich). Segregation of such Fe-rich cumulates thus can enhance the melt oxidation by increasing the Fe^{3+}/ΣFe ratio.

Crystallization at low *P* increases the melt fO_2 by ~1–2 orders of magnitude prior to magnetite saturation, while the final-stage crystallization (at 0.2 kbar) could generate a similar extent of auto-oxidation. Regardless of the initial fO_2, the redox variations of crystallizing melts converge close to QFM at 0.2 and 2 kbar, although further extents of crystallization may cause higher fO_2. However, high-*P* crystallization can increase the melt fO_2 by ~4–8 orders of magnitude. Such extents of auto-oxidation do not appear to be affected by the initial fO_2 but instead increases significantly with *P*. Importantly, the fO_2 variations of MORB glasses, which are associated with thin crusts and low-*P* crystallization, are reproduced well by the modeling results for 0.2 and 2 kbar (Fig. 4b-c and e-f), providing confidence in the robustness of our crystallization models.

4. Magma degassing and crystallization

The above crystallization modeling does not incorporate volatile degassing that may further influence redox systematics. As we discussed in Appendix A.4, previous models for C-H-O-S degassing in magmatic systems are not suitable for the purpose of this study on dynamically and chemically evolving magmatic systems. In this section, we build a new thermodynamic framework to explore how interplays between gas, melt and crystals in dynamically evolving magmatic systems can affect melt fO_2 and gas

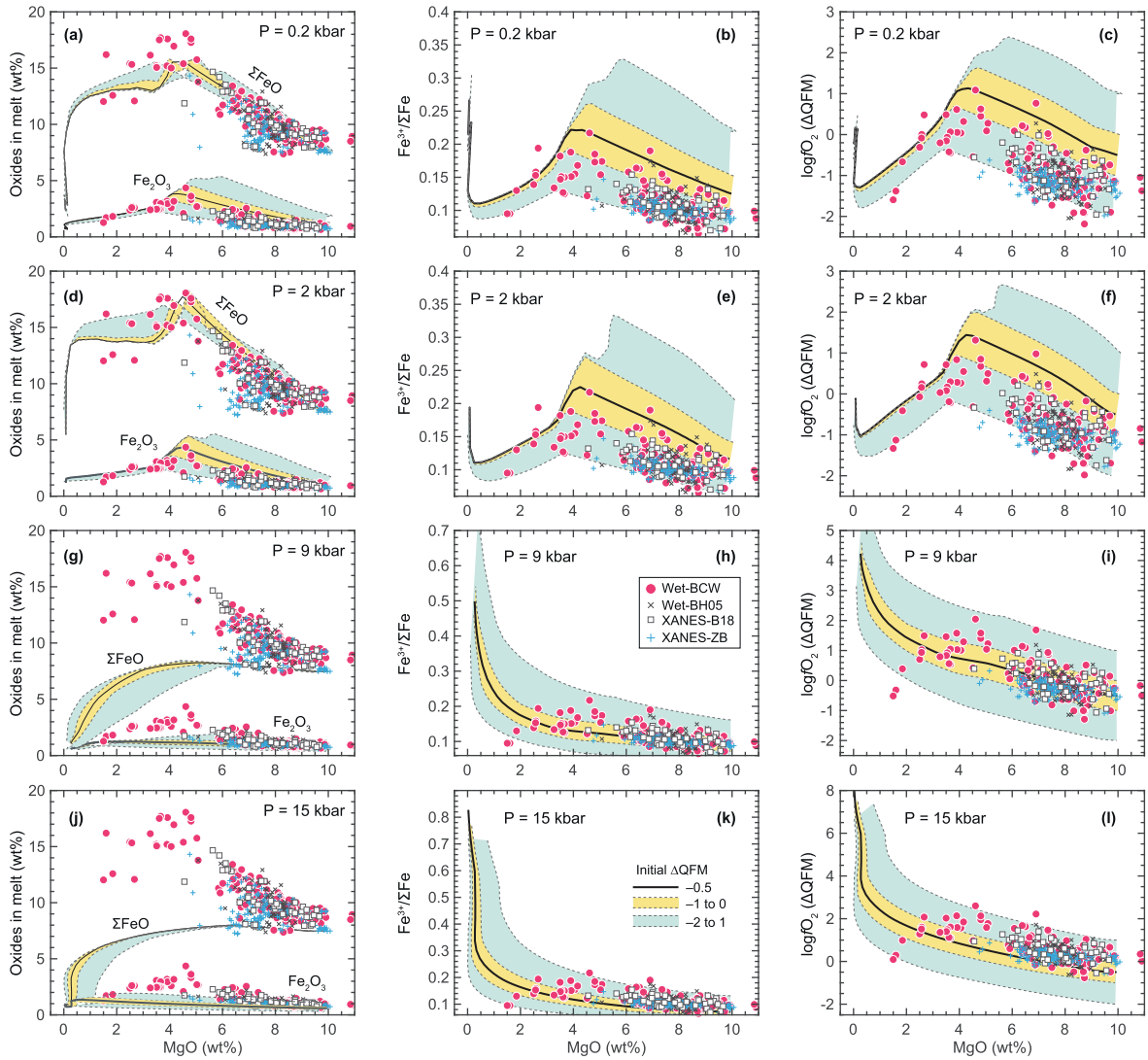

Fig. 4. Modeling results of compositional and redox changes during mantle melt crystallization at 0.2 kbar (a-c), 2 kbar (d-f), 9 kbar (g-i) and 15 kbar (j-l). Plots in the first column show ΣFeO and Fe_2O_3 in the melt against MgO in the melt. Plots in the second and third columns show the model calculated $Fe^{3+}/\Sigma Fe$ ratios and logfO_2, respectively. For comparison, the compiled MORB glasses are also shown using the same symbols as those in Fig. 2. The redox systematics of MORB glasses agree well with the modeling results of crystallization at 0.2 and 2 kbar for the initial fO_2 at ΔQFM = −1.0 ± 0.5.

composition. This framework is coded in a simple MATLAB® program, which is available in the online supplementary materials, to perform Magma and Gas Equilibrium Calculation (i.e., MAGEC) for C-H-O-S bearing systems.

4.1. MAGEC – A new thermodynamic framework for gas–melt coupled systems

This framework incorporates chemical equilibria among different C-H-O-S species in the gas, dissolution of volatiles in the melt, redox reactions between dissolved volatiles and different iron species in the melt, compositional changes of the magmatic system during fractional crystallization, and non-ideal behaviors of gaseous species at elevated *P*. In addition, fO_2 of the system is not fixed at any redox buffers but instead is governed by chemical and redox reactions. The goal of this thermodynamic framework is to simultaneously compute the extent of degassing, post-degassing fO_2, gas composition, and volatile abundances in the melt during crystallization and degassing. The thermodynamic framework of our new model is outlined below.

We considered nine volatile species (i.e., H_2, H_2O, CO, CO_2, CH_4, H_2S, SO_2, S_2, and O_2) in the gas phase. Following previous studies (e.g., Burgisser et al., 2008; Gaillard et al., 2011; Moussallam et al., 2019; Kadoya et al., 2020a), these vapor species maintain homogeneous equilibria through the redox reactions:

$$H_2O = H_2 + \frac{1}{2}O_2, \tag{1}$$

$$CO_2 = CO + \frac{1}{2}O_2, \tag{2}$$

$$CO_2 + 2H_2O = CH_4 + 2O_2, \tag{3}$$

$$\frac{1}{2}S_2 + H_2O = H_2S + \frac{1}{2}O_2, \tag{4}$$

$$SO_2 = \frac{1}{2}S_2 + O_2. \tag{5}$$

Using the NIST-JANAF thermochemical tables (Chase, 1996), we calculated the changes of enthalpy (ΔH) and the equilibrium constants for these reactions at 400–3000 K. The calculated ΔH values indicate that all these reactions are endothermic and thus would consume heat to proceed. Assuming that the vapor phase is an

ideal mixture of real gases, the fugacity of a volatile species is defined as $f = \gamma X P$, where γ is the fugacity coefficient, X is the mole fraction of the volatile species in the vapor phase, and P is the total pressure in bar. To incorporate the gas non-ideality at elevated pressure, we calculated the fugacity coefficients using the equation of state (EOS) of H_2O from Peng and Robinson (1976) and EOS of other vapor species (H_2, CO, CO_2, CH_4, H_2S, SO_2, S_2, and O_2) from Shi and Saxena (1992).

Redox and chemical equilibria between the melt and gas were maintained through the redox reaction between FeO and Fe_2O_3 and the dissolution reactions for various volatile species. The FeO/Fe_2O_3 redox reaction was described using the model of Kress and Carmichael (1991). We noted that several recent models have been calibrated to describe the FeO/Fe_2O_3 redox reaction in silicate melts (e.g., Moretti and Ottonello, 2003; Jayasuriya et al., 2004; Moretti, 2005; Borisov et al., 2018). As the Kress and Carmichael model was implemented in the MELTS program, it was chosen here to be consistent with our crystallization modeling in Section 3. For the dissolution of C and H species in the melt, we adopted the existing solubility models for H_2 (Gaillard et al., 2003), H_2O (Iacono-Marziano et al., 2012), CH_4 (Ardia et al., 2013), CO (Armstrong et al., 2015), and CO_2 (Iacono-Marziano et al., 2012). As S in the melt has two species (S^{2-} and S^{6+}), we used a previous sulfide capacity model (Nzotta et al., 1999) to determine S^{2-} in the melt and then applied a recent sulfur speciation (S^{6+}/S^{2-}) model (Nash et al., 2019) to compute S^{6+} and total S in the melt. To track the saturation of sulfide and sulfate, we calculated SCSS using the model of Fortin et al. (2015) and the S^{6+} content at anhydrite saturation (SCAS) using the model of Zajacz and Tsay (2019). Graphite saturation was monitored using the model of Holloway et al. (1992). Post-degassing fO_2 was determined by balancing the oxygen in the gas phase and that bounded with S and Fe in the melt ($[SO_4]^{2-}$, FeO, and Fe_2O_3). Through mass balance of C, H, O, and S, we also constrained the mass fraction of the total gas in the gas–melt system.

Given the bulk composition of a gas–melt system (C, H, S, major oxides, and $Fe^{3+}/\Sigma Fe$ ratio), the redox/chemical reactions and mass-balance relations outlined above can provide essential constraints (i.e., 18 equations) to simultaneously solve the total of 18 unknowns, including the volatile compositions of the gas (H_2, H_2O, CO, CO_2, CH_4, H_2S, SO_2, S_2, and O_2) and melt (H_2, H_2O, CO, CO_2, CH_4, S^{2-}, and S^{6+}), post-degassing fO_2, and the extent of degassing at a magmatic *P-T* condition. Because these equations are nonlinear, obtaining stable solutions demands reliable numerical solvers and proper initial guesses. To solve these equations, MAGEC adopts a trust-region nonlinear minimization algorithm (Moré et al., 1980) in MATLAB®. At each degassing condition, the numerical solver will be run by up to 50 iterations for different permutations of initial guess values until the resulting solutions reach stable. Although MAGEC is coded in MATLAB, it does not require coding skills for interested users but only needs users to specify the bulk compositions, initial fO_2 (or $Fe^{3+}/\Sigma Fe$ ratio), and *P-T* conditions of degassing as inputs in an Excel® spreadsheet. Users can set different degassing *P-T* conditions to model isobaric or polybaric degassing. The intrinsic uncertainties (= ~20 %) of modeling results in MAGEC are inherited from the solubility models selected from the literature. As MAGEC does not account for error propagations, the modeling results may involve additional uncertainties that need assessment and future improvement through experimental studies.

4.2. Evaluation of MAGEC predictions

Before application of this new framework to complex crustal magmatic zones, it is necessary to test how MAGEC performs in simple systems under laboratory conditions or during magma differentiation. Most laboratory experiments focus on one or two volatile elements (e.g., H, C, and/or S) to investigate the solubilities of their dominant species (e.g., H_2O, CO_2, S^{2-}, and/or S^{6+}) in silicate melts. As MAGEC uses the solubility models from the literature, it is expected to reproduce the experimental data for simple volatile-bearing systems with the same predictive accuracy as the original calibrations. For example, the solubility models of H_2O and CO_2 in MAGEC were taken from Iacono-Marziano et al. (2012) and were calibrated for H_2O- and CO_2-bearing magmatic systems at a wide range of T (= 1100 – 1400 °C), P (= 0.1 – 10 kbar), and composition (SiO_2 = 45 – 63 wt%). Using the reported T, P, and starting compositions, we performed MAGEC calculations for the experiments from Iacono-Marziano et al. (2012). Such calculations assume that the melt-gas systems were completely closed (i.e., no volatile loss or gain) and maintain chemical and redox equilibria during the experimental run. As the fO_2 was not controlled during the run, we used the nominal values (>QFM + 1) suggested by Iacono-Marziano et al. (2012). Despite these assumptions and uncertainties, MAGEC reproduces the measured H_2O and CO_2 contents in the quenched glasses to well within 20% uncertainty (Fig. 5a).

For magmatic systems with multiple C-H-O-S species, however, few experiments are available in the literature to constrain the solubilities of all volatile species. The experimental work of Lesne et al. (2011) is one of the few pioneering studies to directly simulating C-H-O-S degassing from basaltic melts. As Lesne et al. considered relatively oxidized conditions (ΔQFM = ~2 – 3), the dissolved C-H-O-S species in their experiments are dominant by H_2O, CO_2, and S^{6+}. Although their experiments aimed to reach closed-system degassing, Lesne et al. noticed that ~1 wt% H_2O has been added into the experimental charge through oxidation reaction between Fe_2O_3 in the starting material and H_2 from the pressure medium. This is because the metal capsule was open to hydrogen exchange during the run. Given that Lesne et al.'s experiments also included Cl, it is likely that Cl may add additional uncertainties in the solubility determination. As shown in Fig. 5b, MAGEC can predict the dissolved H_2O, CO_2, and total S contents in good agreement with the experimentally measured values, except for those with very low solubilities (<100 ppm). Such agreement is especially salient considering the aforementioned challenges and uncertainties in the experiments.

Using the compiled MORB data, we also evaluated the performance of MAGEC for natural magmas. We used the melt composition, T, P, and $Fe^{3+}/\Sigma Fe$ ratio from our crystallization modeling in Section 3 as the input data for MAGEC calculations. To better match the primitive MORB samples, the initial melt was assumed to crystallize at 2 kbar with 0.15 wt% H_2O, 0.07 wt% CO_2, 0.06 wt% S, and an initial fO_2 at QFM or QFM-1. Because crystallization of cumulate mineral assemblages can preferentially concentrate volatiles in the residual magmas (melt + gas), we corrected the volatile budgets of incremental melts according to the extents of crystallization with an assumed D value (= 0). Fig. 5c-d shows that MAGEC reproduces the differentiation trends of S, H_2O, and CO_2 in MORBs as a function of MgO during crystallization and degassing. This further confirms that MAGEC is a useful quantitative tool to predict the degassing of C-H-O-S from silicate melts.

4.3. Modeling of crystallization and degassing

Using the MAGEC program, we conducted isobaric degassing calculations for magma crystallization at 0.2, 2, 9, and 15 kbar (see detailed results in Table S4). The total C, H, and S contents in the initial magmatic system were adjusted for fractional crystallization by assuming negligible partitioning into crystallizing phases ($D \approx 0$). Results of crystallization modeling from Section 3

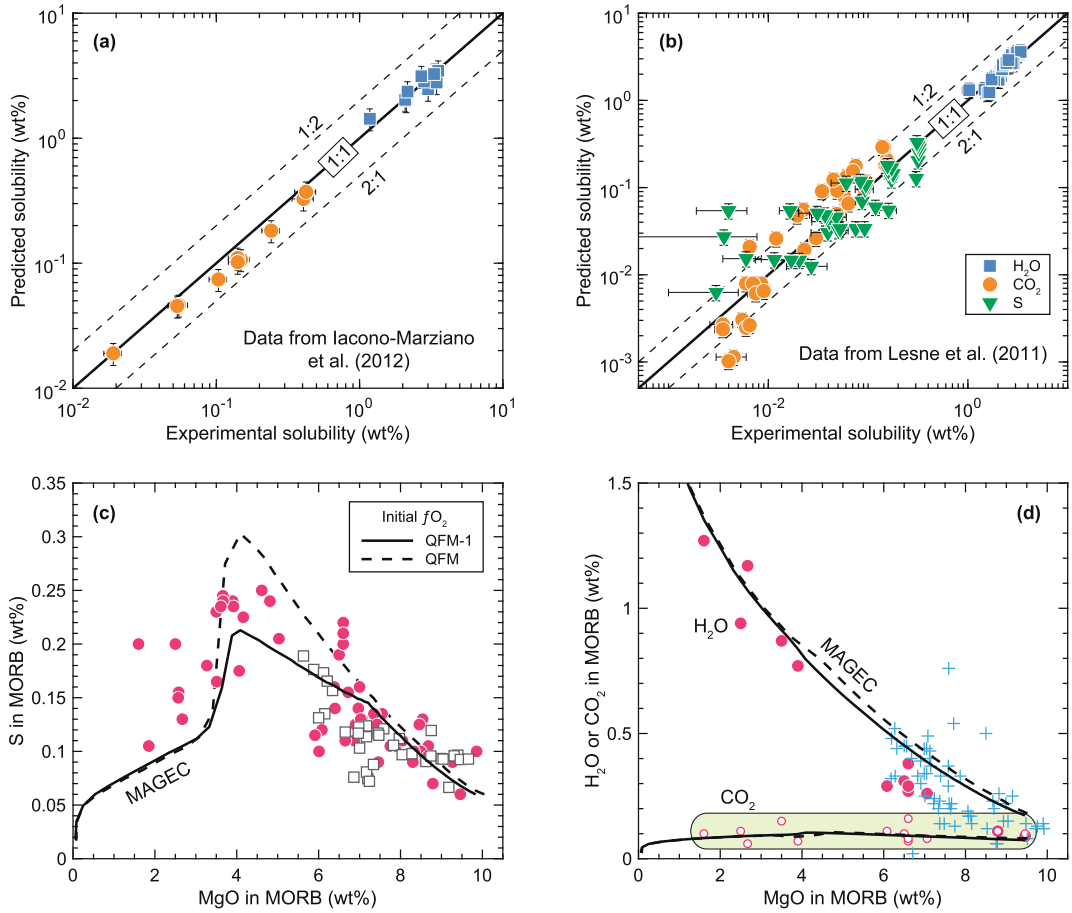

Fig. 5. Independent tests of MAGEC predictions using experimental data (a-b) and MORB glasses (c-d) from the literature. The experimental solubility data of H_2O, CO_2, and S in plots (a-b) are from Iacono-Marziano et al. (2012) and Lesne et al. (2011). Details of the compiled MORB glasses are listed in Table S1. See Fig. 2 for the symbols of MORB samples. Solid and dashed curves in plots (c-d) show MAGEC predictions for C-H-O-S degassing during MORB crystallization.

were taken as the pre-degassing conditions, including *T*, *P*, melt compositions, and pre-degassing fO_2. To be consistent with Section 3, we considered two sets of initial volatile compositions. For each set of initial volatile compositions, two depths were implemented to explore the *P* effect. The low-*P* initial melt (0.2 and 2 kbar) took the volatile composition (2070 ppm H_2O, 697 ppm CO_2, and 1100 ppm S) of the averaged MORB from Arevalo and McDonough (2010), while the high-*P* initial melt (9 and 15 kbar) used the volatile composition (4 wt% H_2O, 1.2 wt% CO_2, and 0.3 wt% S) of the primary arc melt from Plank and Manning (2019). The two sets of volatile compositions were chosen as representative examples of mantle melt differentiation at thin and thick crusts to demonstrate the redox evolution trends.

Fig. 6 shows that magma crystallization can drive volatile degassing at variable depths. This is mainly due to the progressive enrichment of volatile elements in the melt by fractionation of cumulate minerals. As volatile solubilities in the melt generally decrease with *P*, a shallower depth of crystallization could further enhance degassing. With the same initial volatile budget, the highly evolved melt at 2 kbar degasses ~3 wt% gas, while the same melt at 0.2 kbar degasses ~7 wt% gas. The highly evolved melts release similar amounts of gas (~8–9 wt%) at 9 and 15 kbar, although degassing at 15 kbar commences much later than that at 9 kbar. In addition, we observed a small decrease in total gas mass upon melt reduction at low *P*, which is due to the decrease of oxidized species in the gas. This apparent "regassing" phenomenon is quickly overcome by subsequent degassing. Notably, the differences between pre- and post-degassing fO_2 are rather small (within ±0.0025 log units as denoted by the grey regions in Fig. 6), indicating a negligible effect of volatile degassing on the melt redox variation during magma differentiation in the crustal magmatic zone.

Of more interest is how crystallization and degassing influence volatiles in the melt and gas. For illustration, Fig. 7 shows the modeling results for an initial fO_2 at QFM-0.5 (see Table S4 for results of different initial fO_2's). The major volatile species in the melt include H_2O, S^{2-}, S^{6+}, and CO_2. H_2O remains mostly dissolved in the melt and increases with crystallization until saturation is achieved and extensive degassing takes place (Fig. 7a, d, g and j). With a much lower solubility in the melt than H_2O, CO_2 undergoes substantial degassing during crystallization, but it becomes enriched in highly evolved melts at high *P*. This late enrichment can be attributed to the significant depletion of FeO and MgO in highly evolved melts at high *P* (Fig. 4g and j) because CO_2 solubility is inversely correlated with mafic components in the melt according to the solubility law used in our modeling. S in the melt displays a large deviation from the undegassed trend, as it is severely affected by sulfide saturation in addition to degassing. The low-*P* melt meets sulfide saturation shortly after the redox transition (melt MgO = ~4 wt%), when S^{2-} in the melt is increased beyond SCSS through reduction of S^{6+}. With a higher S budget, the high-*P* melt reaches sulfide saturation at the onset of crystallization but later becomes sulfide under-

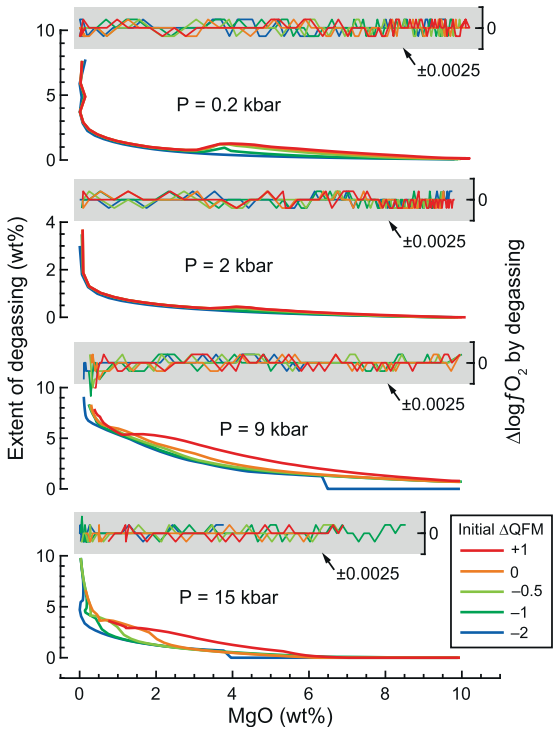

Fig. 6. Variations of the extent of degassing and the difference between post- and pre-degassing fO_2 on a log scale ($\Delta\log fO_2$) as functions of MgO in the melt from thermodynamic modeling of crystallization and degassing at variable depths (0.2–15 kbar) and initial fO_2 (ΔQFM = −2 to +1). Grey areas denote the ±0.0025 variations of $\Delta\log fO_2$. The small fluctuation of $\Delta\log fO_2$ is likely due to numerical errors from the nonlinear minimization algorithm.

saturated because extensive auto-oxidation decreases S^{2-} concentrations to levels below SCSS by oxidation of S^{2-} and degassing of the oxidized S species.

The gas compositions also show systematic variations during crystallization and degassing at variable depths (Fig. 7b, e, h, and k). Although CO_2 reaches saturation earlier than H_2O, the early degassed CO_2 allows continuous H_2O degassing by reducing H_2O partial *P*. The reduced C-H species (CO, CH_4, and H_2) have diverse behaviors during crystallization. Regardless of *P*, CO in the gas increases during melt reduction and decreases during melt oxidation. A similar redox effect is also evident in H_2 and CH_4, except for those during early melt oxidation at low *P* (Fig. 7b and e). According to the redox reactions in Eqs. (1)–(3), the reduced C-H species in the gas are negatively affected by fO_2 but are positively correlated with CO_2 and H_2O. The increase of H_2O apparently interferes with the redox effect during early-stage crystallization at low *P*, resulting in the slight increase of H_2 and CH_4 in the gas.

Among the three S species (i.e., SO_2, S_2, and H_2S) in the gas, SO_2 is the primary one under sulfide-undersaturated conditions, while H_2S is the dominant species at sulfide saturation. Due to the redox effect (cf. Eqs. (4)–(5)), SO_2 in the gas increases with melt oxidation and decreases with melt reduction. S_2 generally follows SO_2 in the low-*P* gas but is strongly affected by H_2S in the high-*P* gas. However, H_2S manifests complex behaviors that do not always comply with the redox effect. We noted that H_2S in the low-*P* gas mimics H_2, although the former is more abundant than the latter by about one order of magnitude. This indicates that H_2S is also controlled by H_2O in addition to fO_2 (cf. Eq. (4)). The increase of H_2O in the gas appears to control the increase of H_2S during melt oxidation at variable depths. At high *P*, the final-stage auto-oxidation becomes too intensive to be overwhelmed by the effect of H_2O and accordingly yields a rapid decrease of H_2S. As the primary S species at sulfide saturation, H_2S in the gas is also affected by sulfide separation. Specifically, sulfide saturation suppresses the increase of H_2S during melt reduction at low *P* and during melt auto-oxidation at high *P*.

5. Implications for atmospheric oxygenation

5.1. Reducing power of magmatic gases

The efficiency of atmospheric O_2 consumption by the magmatic gas depends on the relative abundances of reduced volatile species. For a particular volatile element (e.g., H, C, or S), the molar ratio between its reduced and oxidized species (e.g., H_2/H_2O, CO/CO_2, or H_2S/SO_2) can be used to measure its reducing capacity. As determined by the redox reactions in Eqs. (1)–(2), and (4)–(5), the reducing capacities of H, C, and S in the gas are inversely correlated with fO_2 (Fig. 7c, f, i, and l). Due to the large stoichiometry coefficient of O_2 in Eqs. (4)–(5), the reducing capacity of S appears to be more sensitive to fO_2. During late-stage crystallization, when most intensive degassing takes place, the reducing capacities of H, C and S in high-*P* gases are less than those of the same elements in low-*P* gases by ~3–8 orders of magnitude. This suggests that high-*P* magmatic gases are much less effective O_2 sinks.

We note that individual elements may not be representative for the bulk gas, as their reducing capacities can vary by several orders of magnitude (e.g., H_2S/SO_2 vs CO/CO_2 at 2 kbar; Fig. 7f). To better describe the efficiency of O_2 removal by the bulk magmatic gases, we defined an index of reducing power (R_p),

$$R_p = \frac{0.5H_2 + 0.5CO + 2CH_4 + 1.5H_2S + 2S_2}{H_2O + CO_2 + SO_2}, \quad (6)$$

where gas species are in mole fraction. The numerator is the total O_2 to be consumed by reduced gas species, the coefficients of which indicate their relative contributions. The denominator is the sum of oxidized gas species as a normalization factor. R_p is distinct from the *f* value defined by Holland (2002) or other analogs of Holland's *f* value (e.g., Kump and Barley, 2007; Kasting, 2013; Kadoya et al., 2020b), because it is independent of organic or pyrite burial. Applying Eq. (6) to the results shown in Fig. 7, we found that low-*P* gases show nearly constant R_p on a log scale across the entire interval of crystallization and degassing. The extremely high H_2S/SO_2 ratios in the late-stage gases at low *P* do not appear to affect R_p, because SO_2, the controlling factor for high H_2S/SO_2 ratios, becomes a trace component in the bulk gas. At high *P*, R_p's are also nearly constant during early-stage crystallization but begin to decrease abruptly at later stages (Fig. 7i and l). As to the extensive degassing from late-stage crystallization, R_p's at high *P* are ~2–5 orders of magnitude lower than those at low *P*, indicating that gases released from deeper magma chambers are significantly less efficient in O_2 removal.

The above finding was based on the modeling results for an initial fO_2 at QFM-0.5. We further tested this notion for initial melts from more reduced and more oxidized mantle sources (ΔQFM = −2, −1, 0, and +1). As shown in Fig. 8a-b, although R_p's at low *P* are affected by the initial fO_2 during early-stage degassing, they converge upon sulfide saturation when H_2S becomes the dominant reduced gas species. At high *P* (Fig. 8c-d), R_p's for variable initial fO_2 are similar during early-stage degassing at sulfide saturation; however, during extensive auto-oxidation, when sulfide becomes undersaturated, all reduced gas species undergo a significant decrease that leads to a dramatic decline of R_p. In addition, more oxidized initial melts at high *P* appear to drive earlier declines of R_p. This is because higher fO_2 can enhance the conversion of S^{2-} to S^{6+} in the melt, resulting in a shorter interval of sulfide saturation. Notably, R_p's from extensive degassing at 15 kbar are lower

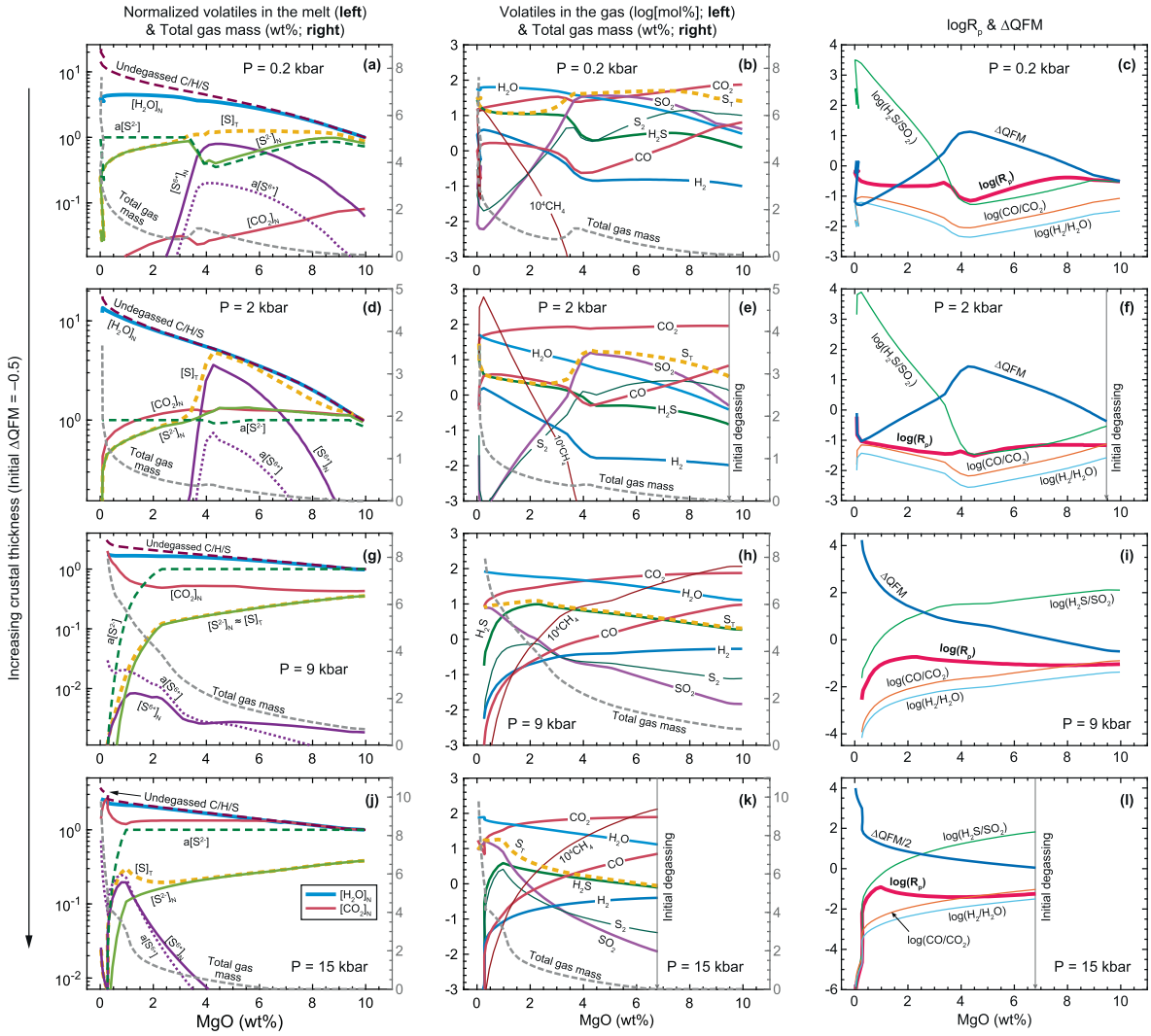

Fig. 7. Modeling results of volatile compositions in the melt and coexisting gas during crystallization and degassing of a mantle-derived melt with the initial fO_2 of QFM-0.5 at 0.2 kbar (a-c), 2 kbar (d-f), 9 kbar (g-i) and 15 kbar (j-l). Plots in the first column show major volatile species (H_2O, CO_2, S^{2-}, S^{6+}, and total S) dissolved in the melt. To highlight the volatile loss by degassing and/or sulfide separation, these dissolved volatile species are normalized by the concentrations of H_2O, CO_2, or total S in the initial melt (i.e., $[H_2O]_N$, $[CO_2]_N$, $[S^{2-}]_N$, $[S^{6+}]_N$, and $[S]_T$). The activity of S^{2-} ($a[S^{2-}]$) in the melt is defined by the ratio between the dissolved S^{2-} in the melt and SCSS, while that of S^{6+} ($a[S^{6+}]$) is defined by the ratio between the dissolved S^{6+} in the melt and SCAS. Plots in the second column show the compositions of magmatic gases at variable depths on a log scale. For references, the extents of degassing (grey dashed curves) are shown on the right axis. Plots in the third column show reducing capacities of individual volatile elements (i.e., molar ratios of H_2/H_2O, CO/CO_2, and H_2S/SO_2), the reducing power indices of the bulk gas (R_p), and the fO_2 of the gas–melt system (ΔQFM) on a log scale.

than those at 2 kbar by at least ~5 log units despite variation in initial fO_2 (Fig. 8d). R_p's from large extents of degassing at 9 kbar are also less than those at 2 kbar by ~2–5 orders of magnitude except those for the highly reduced initial fO_2 (QFM-2; Fig. 8c). Given that Earth's upper mantle fO_2 is at least above QFM−1.2 since the Archean (e.g., Lee et al., 2003; Aulbach and Stagno, 2016; Nicklas et al., 2019), the calculated R_p's confirm that high-P magmatic gases are much less effective O_2 sinks regardless of mantle fO_2.

5.2. Effects of polybaric degassing and gas transport

Our discussion above assumed instantaneous release of gases from the crustal magmatic zone to the atmosphere. A more realistic scenario should also account for the gas transport. In this section, we use thermodynamic modeling to assess to what extent the decompressional transport of gases can affect their R_p's in the atmosphere. Here, we considered two endmember cases: (1) minimum interactions between the gas and surrounding magma and (2) complete gas–melt equilibrium during transport from the dynamically evolving magmatic zone to surface.

As noted by Moussallam et al. (2019), global volcanic gases show a negative correlation between fO_2 and T, suggesting closed system behaviors during gas cooling. Due to the lack of constraints on the P of last-stage gas–melt equilibration, it remains unclear to what extent the fO_2 variation of global volcanic gases is affected by decompression. Using the redox reactions in Eqs. (1)–(5) and the NIST-JANAF thermochemical tables (Chase, 1996), we performed thermodynamic calculations for closed gas systems. The initial gas compositions (Table S5) were taken from the last steps of modeling results in Section 4.3 for an initial fO_2 at QFM-0.5. These gases were allowed to decompress isothermally or with cooling (cf. solid vs dashed curves in Fig. 9a). Results of thermodynamic calculations show that isothermal decompression decreases the gas fO_2 by ~2–4 orders of magnitude because of the P effect (cf. solid curves in Fig. 9b). Considering the ideal gas behavior for simplicity, the decrease of P at constant T is equivalent to the increase of volume (V). According to Le Chatelier's principle, the increase of

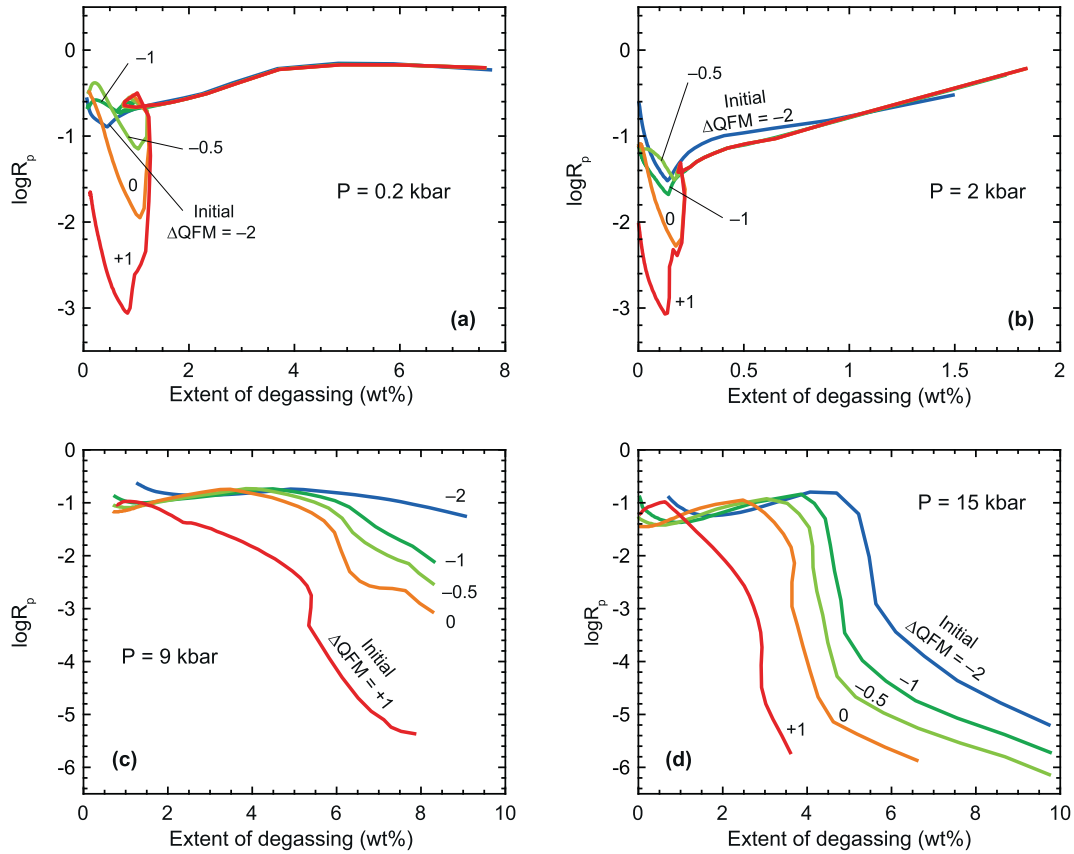

Fig. 8. Plots showing the reducing power indices (R_p) of magmatic gases as a function of the extent of degassing during crystallization at 0.2 kbar (a), 2 kbar (b), 9 kbar (c), and 15 kbar (d). Curves of different colors indicate results for variable redox conditions of the initial melt.

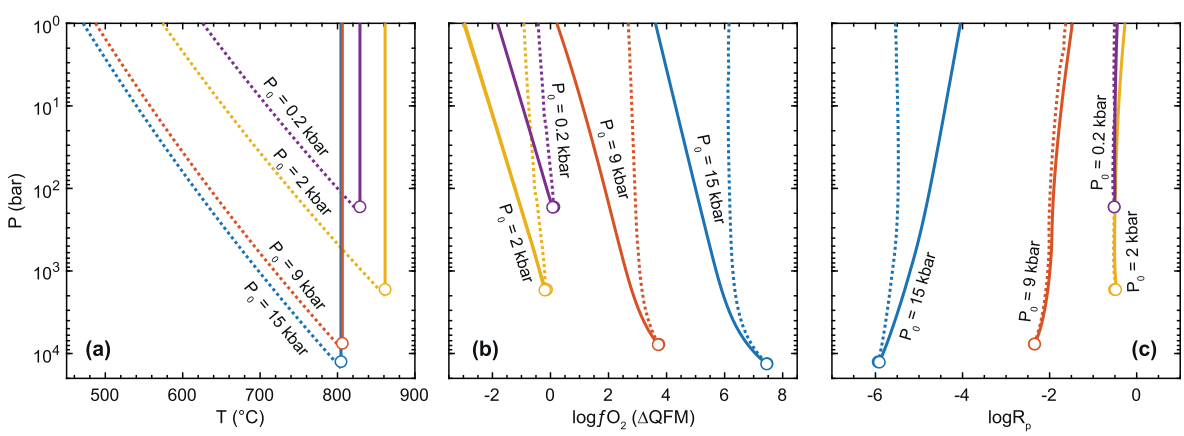

Fig. 9. Modeling results of decompressional transport of magmatic gases from the crustal magmatic zone to surface in closed systems. Plots (a-c) show the profiles of T, $\log fO_2$, and $\log R_p$ of magmatic gases from variable initial pressures (P_0) of degassing. Solid and dashed curves indicate constant and variable T, respectively, during decompression. The bulk gas compositions and initial conditions of degassing are listed in Table S5.

V drives the equilibrium towards the side with more moles of gases. Hence, during isothermal decompression, the equilibrium reactions of Eqs. (1)–(5) will shift towards the direction of increasing reduced species (i.e., H_2, CO, and S_2), leading to a lower fO_2. A larger extent of decompression yields a more significant decrease of fO_2. Given the different P intervals, the gas R_p's from low-P magmas show marginal decreases, but those from high-P magmas decrease by ~1–2 orders of magnitude (cf. solid curves in Fig. 9c). Upon release to the surface, gases from high-P magmas still show considerably lower R_p's than those from low-P magmas (i.e., ~2–4 orders of magnitude).

Because the gas equilibrium reactions (Eqs. (1)–(5)) are all endothermic, cooling of the closed gas system favors reverse reactions that can increase the oxidized species (i.e., H_2O, CO_2, SO_2, and S_2). Hence, cooling should yield a higher fO_2, which is opposite the effect of decompression. To examine the combined effect of cooling and decompression, we described the T of the closed gas system as $T = T_0\,(P/P_0)^{1.04}$, where subscript "0" denotes the initial value prior

to degassing and the exponent "1.04" is arbitrarily selected to describe the extent of cooling. This cooling profile (cf. dashed curves in Fig. 9a) is merely for demonstration because simple formulations are not available for adiabatic decompression of multicomponent gases. Results of thermodynamic calculations show that for the considered T intervals, the combined effects of cooling and decompression only reduce the fO_2 of low- and high-T gases by less than one order of magnitude (Fig. 9b). The extents of fO_2 decreases are ~1–2 orders of magnitude smaller than those from isothermal decompression, resulting in negligible variations of R_p's (Fig. 9c). Given the competing effects of P and T, cooling of closed gas systems during decompressional transport should not overturn the distinct R_p's between low- and high-P gases.

It is also possible that the exsolved gases are largely in equilibrium with the surrounding magmas during decompressional transport from the deep crustal magmatic zone to surface. To examine how gas R_p's change in this case, we simulated polybaric degassing using the MAGEC program (Section 4.1). As we discussed above for closed gas systems, cooling is expected to increase gas fO_2, but for polybaric degassing this is difficult to model because cooling can induce crystallization and modify the magma composition. For simplicity, we considered polybaric degassing at constant T. The evolved magmas (i.e., melt + gas) at ~1000 °C from isobaric crystallization at variable depths (Section 4.3) were taken as the initial compositions of the gas–melt systems. Details of the initial compositions and T–P–fO_2 conditions are listed in Table S6.

Results of MAGEC calculations show that polybaric degassing yields different consequences on the redox variations of gases from low- and high-P magmas (Fig. 10). During polybaric degassing, the low-P gases (Fig. 10a-b) show a slight increase in fO_2 followed by a decrease of fO_2. The overall fO_2 variations of low-P gases are about one order of magnitude. However, the high-P gases (Fig. 10c-d) show a continuous decrease of fO_2 by ~2 orders of magnitude. The increase of fO_2 during early degassing of low-P magmas (Fig. 10a-b) can be attributed to the release of H_2O from the melt, as it effectively reduced the H_2/H_2O ratios in the gas. When major compositions of the gases become steady (i.e., nearly constant H_2O/CO_2 ratios), P will become the controlling factor for the gas

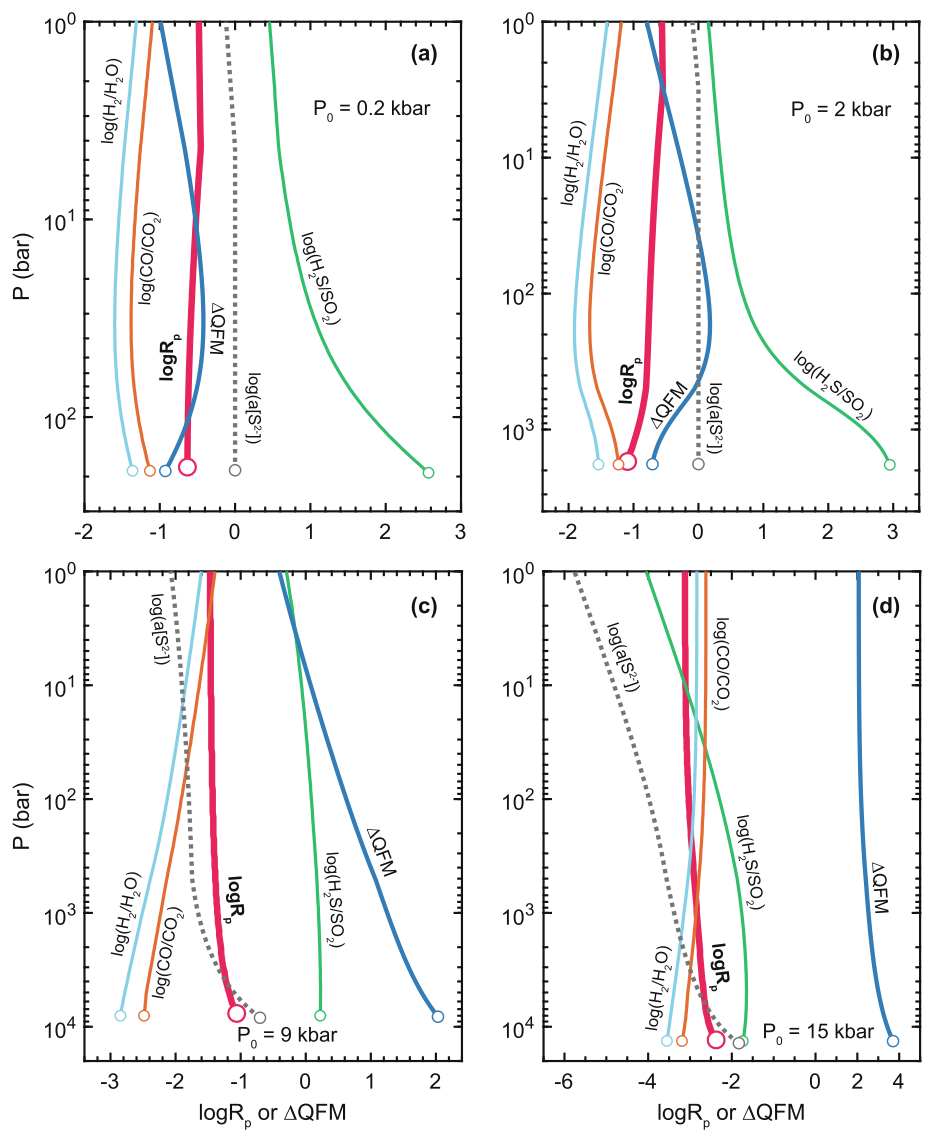

Fig. 10. The reducing power indices (R_p) and redox conditions (ΔQFM) of magmatic gases from modeling results of polybaric degassing at initial pressures (P_0) of 0.2 kbar (a), 2 kbar (b), 9 kbar (c), and 15 kbar (d). The evolved magmas at ~1000 °C during isobaric crystallization with an initial fO_2 of QFM-0.5 were selected as the bulk magma compositions (see details in Table S6). The reducing capacities of individual volatile elements (i.e., molar ratios of H_2/H_2O, CO/CO_2, and H_2S/SO_2) and the activity of S^{2-} ($a[S^{2-}]$) are also shown for comparison. See Fig. 7 for the definition of $a[S^{2-}]$.

redox evolution. As discussed above for closed gas systems (Fig. 9), decompression reduces the fO_2 by shifting the redox reactions towards the direction with more moles of gases. Given the increase of H_2/H_2O and CO/CO_2 ratios with the decrease of P, the P effect is clearly the main driver for the decrease of fO_2 during late-stage degassing of low-P magmas and across the entire degassing intervals of high-P magmas.

It is interesting that the H_2S/SO_2 ratios of the gas do not follow an expected inverse correlation with fO_2 during polybaric degassing (Fig. 10). Instead, they generally decrease with the decrease of P, especially at shallower depths ($P < \sim 2$ kbar) when the major gas compositions approach steady. In the low-P melts, sulfide remains saturated until $P < \sim 5$ bar and therefore can continuously supply S in the gas while buffering S in the melt at SCSS. On the other hand, the high-P melts do not meet sulfide saturation during polybaric degassing. Without a buffer (sulfide or sulfate), S preferentially partitions into the gas during early degassing of high-P melts and subsequently becomes diluted with further extents of degassing. Such distributions of S between the gas and melt do not appear to affect the H_2S/SO_2 ratios of the gas. We note that H_2S/SO_2 should establish equilibrium with the primary redox pairs (i.e., H_2/H_2O, and CO/CO_2) in the gas through:

$$SO_2 + 3H_2 = H_2S + 2H_2O, \quad (7)$$

$$SO_2 + 3CO + H_2O = H_2S + 3CO_2. \quad (8)$$

Equation (7) was also noted in many previous studies (e.g., Moretti et al., 2003; Aiuppa et al., 2005; Gaillard et al., 2011). Given that both reactions (Eqs. (7)–(8)) have more moles of gases on the left side, they will proceed towards the left when P decreases, yielding a lower H_2S/SO_2 ratio but higher H_2/H_2O and CO/CO_2 ratios. Note that this indicates an important redox competition between S and the primary components (i.e., H and C) in the gas. During polybaric degassing of low-P melts, the added S in the gas will preferentially turn into oxidized S species and thus can further enhance the decrease of H_2S/SO_2 ratios in the gas. Because of the redox competition between H_2S/SO_2 and the primary redox pairs (i.e., H_2/H_2O, and CO/CO_2), polybaric degassing only results in small variations of R_p's for both low- and high-P gases (i.e., less than one order of magnitude). Particularly, the gas R_p's become slightly decreased during shallow degassing of low-P melts ($P < \sim 5$ bar; Fig. 10a-b) and across the entire degassing intervals of high-P melts (Fig. 10c-d). This is due to the dilution of S in the gas by degassing of other volatile species when the melts are undersaturated with sulfide.

5.3. Effects of crustal thickness

The substantial decrease of R_p's from low- to high-P magmatic gases reveals a profound effect of the average depth of crystallization on the efficiency of O_2 removal from the atmosphere (Fig. 11). As dynamically evolving magmatic systems underneath thin crusts are governed by low-P crystallization, they can generate substantial emission of high-R_p magmatic gases to enhance O_2 removal

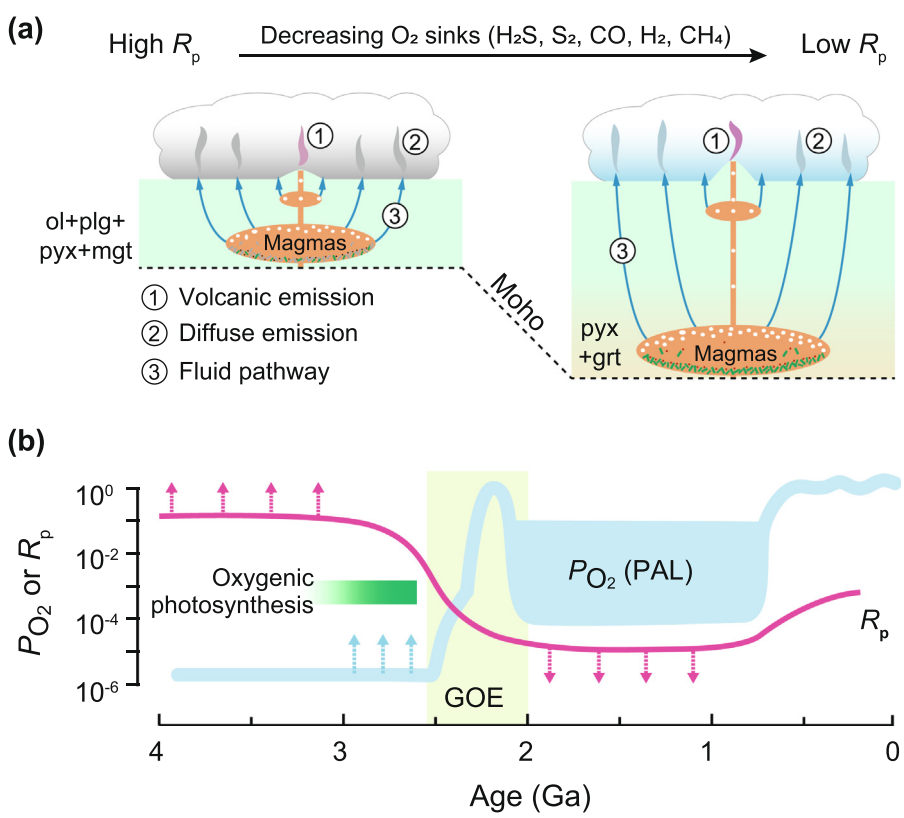

Fig. 11. Magmatic gas emission and its impacts on atmospheric oxygenation. Plot (a) is a schematic drawing for gas emission from solidifying magmas during igneous crust formation. ① and ② denote volcanic and diffuse emission, respectively, and ③ indicates the high-permeability channels or fractures as fluid pathways. Moho denotes the crust-mantle boundary. The magmatic system underneath a thin crust fractionates mostly olivine (ol), plagioclase (plg), pyroxene (pyx) and magnetite (mgt) to form gabbroic cumulates, whereas those underneath a thick crust form high-P cumulate rocks dominated by pyx, garnet (grt), and/or amphibole. Crystallization of different mineral assemblages control the redox evolution of the melts, volatile degassing, and the gas compositions. The reduced gas species (H_2, CO, CH_4, S_2 and H_2S) are effective O_2 sinks by directly consuming atmospheric O_2 through photochemical reactions. Plot (b) shows the variation of the atmospheric O_2 level (Lyons et al., 2014) through the Earth history. Blue curve and box in (b) indicate the atmospheric O_2 partial P (PO_2) relative to the present atmospheric level (PAL). Green bar denotes the initiation of oxygenic photosynthesis. R_p is schematically inferred from the active crustal thickness (Dhuime et al., 2015). Red and blue dashed arrows indicate the possible variations of R_p and PO_2. An increase in active crustal thickness (i.e., during crust formation) through time leads to a decrease in R_p and hence the sink of O_2, allowing atmospheric O_2 levels to rise.

from the atmosphere. However, magma differentiation underneath thick crusts are subject to high-P crystallization and therefore can drive extensive degassing of low-R_p magmatic gases with much less efficiency in O_2 consumption. This establishes a possible causal link between crustal thickness during magmatism and the efficiency of magmatic gases in O_2 removal. Independent tests for this causal link require comprehensive analyses of major and minor volatile species in magmatic gases from thin and thick crusts. Applying Eq. (6) to the compilation of Kump and Barley (2007), we calculated R_p's for gases from different tectonic settings of modern Earth. R_p's of the submarine gases from the ocean floor ($\log R_p$ = −0.67 – 0.59, averaged at −0.02 ± 0.20) are about two orders of magnitude greater than those of subaerial gases from convergent margins and continental rift zones ($\log R_p$ = −3.60 to −0.43, averaged at −1.93 ± 0.77). The Moho depths of the latter are generally greater than those of the former by ~10–50 km (e.g., Mooney et al., 1998; Szwillus et al., 2019). Considering the uncertainties due to variable extents of meteoric water alteration and/or melt-wall rock interaction, the large difference in R_p's of gases from thin and thick crusts is statistically significant, supporting the results of our thermodynamic modeling. Hence, any secular variations of crustal thickness during global crust formation will impact the role of magmatic gases as an O_2 sink and thus influence the evolution of long-term atmospheric O_2 levels across Earth's history. We stress that this point must be considered in any models for understanding the long-term evolution of atmosphere oxygenation.

There is a growing body of evidence indicating that the first massive pulse of continental crust formation and thickening occurred during the late Archean to early Proterozoic and that such crust production was associated with thick crusts (Durrheim and Mooney, 1991; Keller and Schoene, 2012; Dhuime et al., 2015; Tang et al., 2020). For instance, global seismic models (Durrheim and Mooney, 1991; Keller and Schoene, 2012) suggest that the crusts generated during the Paleoproterozoic are thicker than those formed in the mid-Archean by ~10 km. Similarly, various geochemical proxies require the late Archean-Paleoproterozoic juvenile crusts to have been thick (Dhuime et al., 2015; Tang et al., 2020). If the inferred changes in active crustal thickness are true, the extensive crustal thickening and growth from the late Archean to Paleoproterozoic might suggest a sharp increase in the depth of magma differentiation over the same time interval, leading to the production of low-R_p magmatic gases. Combined with the fact that magma differentiation in thick crusts also segregates most of the reduced Fe and S species as cumulates in the lower crust (Lee et al., 2016), the first major pulse of continental crust formation and thickening may have been accompanied by a significant decrease in the reductive capacity of magmatic gases and Earth's surface, allowing atmospheric O_2 to rise to a steady level for the first time.

6. Conclusion

Using petrological observations and new geochemical modeling, we demonstrated that the redox states of dynamically evolving magmatic systems are controlled by the nature of magma differentiation, including the mineralogy of cumulates, the extent of solidification, and the depth of crystallization. Because Fe^{3+} is generally more incompatible than Fe^{2+} in silicate minerals, magma differentiation tends to increase the $Fe^{3+}/\Sigma Fe$ ratio and results in auto-oxidation of the melt. At low P, this auto-oxidation eventually transitions into auto-reduction by when magnetite saturates in moderately evolved melts; however, at high P, this auto-oxidation becomes progressively enhanced mainly due to preferential Fe^{2+} depletion from the melt by garnet fractionation. Isobaric degassing of C-H-O-S volatiles from the solidifying magmas at variable depths does not impact melt redox variation, but the latter can fundamentally affect the compositions of magmatic gases and their reducing powers for atmospheric O_2 removal. Upon extensive degassing, high-P magmatic gas emission has a much smaller reducing power than the low-P ones, because of the extensive melt auto-oxidation at elevated P. The reducing powers of these magmatic gases are insensitive to gas decompressional transport in closed systems or polybaric degassing in gas–melt equilibrium systems. As high-P magmatic gases can only be generated from thick crusts, our findings indicate that thick active crust formation can substantially decrease the efficiency of O_2 consumption by magmatic gases, which may have important implications for the atmospheric O_2 rise in the Paleoproterozic.

Data availability

All data supporting the findings of this study can be found in the supplementary material and the cited references.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

A.1. Data correction and oxygen fugacity calculation

We corrected the XANES analyses in Zhang et al. (2018) and Birner et al. (2018) following the expression, $[Fe^{3+}/\Sigma Fe] = ([Fe^{3+}/\Sigma Fe]_o + 0.016 - 0.065)/(1 - 0.065)$, where the subscript "o" indicates the original data. This formula was modified from the expression in Berry et al. (2018) by incorporating the correction of Cottrell and Kelley's XANES calibration used in Zhang et al. (2018) and Birner et al. (2018). The wet chemical analyses were then adjusted according to the mean deviations from the XANES analyses (Birner et al., 2018; Zhang et al., 2018) for primitive MORB glasses (MgO ≥ 8 wt%). The data from Bézos and Humler (2005) were treated separately, as they used a different wet chemical method. The averaged $Fe^{3+}/\Sigma Fe$ ratio (= 0.096 ± 0.012) of primitive MORBs from XANES analyses is smaller than that (= 0.116 ± 0.021) from Bézos and Humler (2005) by 0.020 and is greater than that (= 0.068 ± 0.017) from other wet chemical studies by 0.028.

In addition to the $Fe^{3+}/\Sigma Fe$ ratio, the model of Kress and Carmichael (1991) (noted as KC91 model hereafter) also needs information of T, P, and melt compositions. In our compilation, 290 samples were reported with major element compositions (denoted as M290), while 123 samples (denoted as L123) lacked detailed composition information except for MgO and FeO. For the M290 samples, we applied the magma thermometer of Sun and Dasgupta (2020) to calculate the liquidus T's at 2 kbar and then obtained the fO_2 using the KC91 model. To calculate T's for the L213 samples, we used the empirical relationship between T's

and MgO contents of the M290 glasses, $10^4/T$ (K) = −0.73 ln[MgO (wt%)] + 8.45. This simple equation reproduces T's with a 1σ error of ±9 K (Fig. A.1a). Among the L123 glasses, 5 samples do not have any additional compositional information, and 118 were reported with the compositional term ($\sum dX$) of the KC91 model (N = 47) or an earlier model (Kilinc et al., 1983) (noted as K83 model; N = 71). As shown in Fig. A.1b, the ratios between $\sum dX_{K83}$ and $\sum dX_{KC91}$ for the M290 samples show a strong inverse correlation with ΣFeO contents, following a simple expression, $(\sum dX_{KC91})/(\sum dX_{K83})$ = −0.05 ∑FeO (wt%) + 0.81. This expression was then used to estimate $\sum dX_{KC91}$ for the 71 L123 samples based on their $\sum dX_{K83}$. Overall, we were able to calculate fO_2 for 408 samples using the KC91 model.

A.2. Ferric iron partitioning between magnetite and melts

We compiled 445 high-T experiments with coexisting magnetite and silicate melts from the literature (see Table S2 for references). These experiments cover a broad range of T (= 747 – 1325 °C), P (= 1 – 7046 bar), and controlled fO_2 (ΔQFM = −1.2 to +7.6). Silicate melts from these experiments vary from basaltic to rhyolitic compositions (SiO_2 = 43 – 79 wt%) with a wide range of Mg# (= 0 – 68). To obtain a self-consistent dataset, we followed the previous approach and re-calculated Fe_2O_3 abundances in the magnetite crystals and silicate melts from the compiled experiments. Specifically, Fe_2O_3 in magnetite was constrained by stoichiometry, and Fe_2O_3 in the melt was calculated using the model of Kress and Carmichael (1991). Magnetite-melt D's of Fe_2O_3 were then obtained by taking the ratios of Fe_2O_3 (wt%) in magnetite and coexisting melts. Table S2 lists the details of the experimental data together with the calculated D's of Fe_2O_3.

As shown in Fig. A.2a-c, magnetite-melt D's of Fe_2O_3 are significantly greater than one and vary by ∼1.5 orders of magnitude (D = 5–159) as a function of melt composition (e.g., SiO_2 and FeO) and crystal chemistry (e.g., Al_2O_3). Through a linear least-squares regression analysis, we found that magnetite-melt D's of Fe_2O_3 can be described by the following equation,

$$\ln D = -0.56(\pm 0.15) - 0.74(\pm 0.03)\ln X_{Fe} + 2.48(\pm 0.34)X_{Si} - 2.61(\pm 0.39)Al^{3+}, \quad (A.1)$$

where X_{Si} and X_{Fe} denote the cation fractions of Si and total Fe (in mole) in the melt, Al^{3+} indicates the cation fraction of Al in magnetite, and numbers in parentheses are 1σ errors of the coefficients derived from the regression analysis. This equation can reproduce the D values to well within a factor of two (Fig. A.2d). Notably, fO_2 has no direct influences on the partitioning of Fe_2O_3 between magnetite and melts.

A.3. Effect of amphibole crystallization on the redox evolution of high-P melts

Amphibole is a common hydrous mineral in high-P cumulate rocks. Because Fe_2O_3 is moderately compatible in amphibole (e.g., King et al., 2000), the lack of amphibole-rich cumulate assemblages from our crystallization modeling (Section 3) can potentially lead to overestimation of melt auto-oxidation during high-P crystallization. To assess the potential effect of amphibole on Fe_2O_3 fractionation, we conducted crystallization simulations at 9 and 15 kbar using a higher bulk D value (= 0.48) for Fe_2O_3. This bulk value D was calculated by assuming 30% of amphibole in the cumulate assemblage. Unfortunately, there are no rigorous models available in the literature to quantify the partitioning of Fe_2O_3 between amphibole and melt. For demonstration, we took the D value of Fe_2O_3 for amphibole from that of ΣFeO (D = 1.37) in the experiment of King et al. (2000) under highly oxidized condition (∼QFM + 5). The D value of Fe_2O_3 for other silicate minerals is identical to that used in Section 3. As shown in Fig. A.3, the higher bulk D value for amphibole-rich cumulate assemblages yields slightly smaller extents of melt auto-oxidation than that (D = 0.1) of amphibole-free assemblages. The resulting increase of melt fO_2 is about 3.5–6.5 orders of magnitude and remain significantly greater than that from low-P crystallization. This suggest that crystallization of amphibole-rich cumulate assemblages does not appear to affect the first-order redox variation during high-P crystallization.

A.4. Previous models for C-H-O-S degassing from silicate melts

Several degassing models have been constructed for C-H-O-S bearing magmatic systems in previous studies, such as D-Compress (Burgisser et al., 2008), SolEx (Witham et al., 2012), and the models of Moretti et al. (2003; M03), Scaillet and

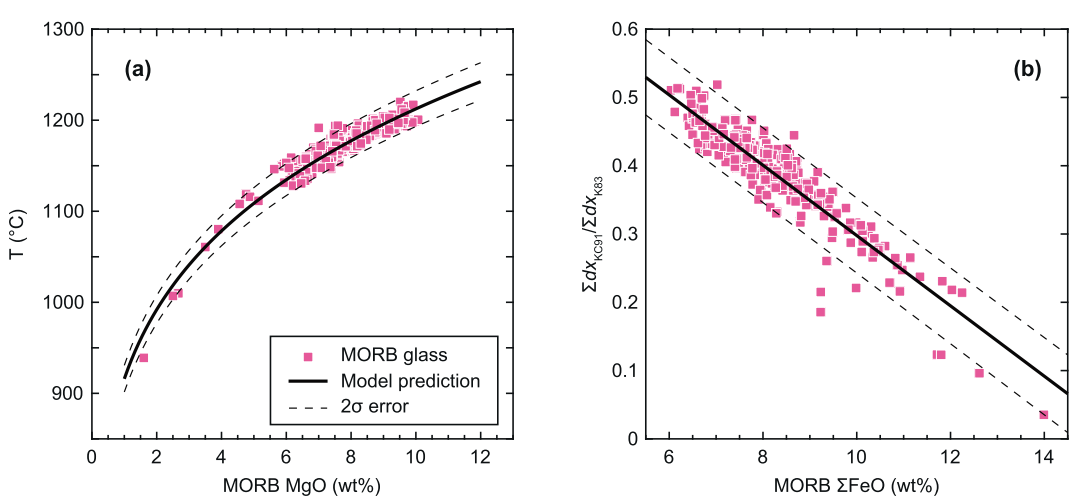

Fig. A1. Empirical estimations of liquidus T and ΣdX_{KC91} for MORB glasses. Plot (a) shows the correlation between liquidus T and MgO of the MORB glasses that were reported with major element compositions. Plot (b) shows the ratios between ΣdX_{KC91} and ΣdX_{K83} as a function of ΣFeO in the MORB glasses.

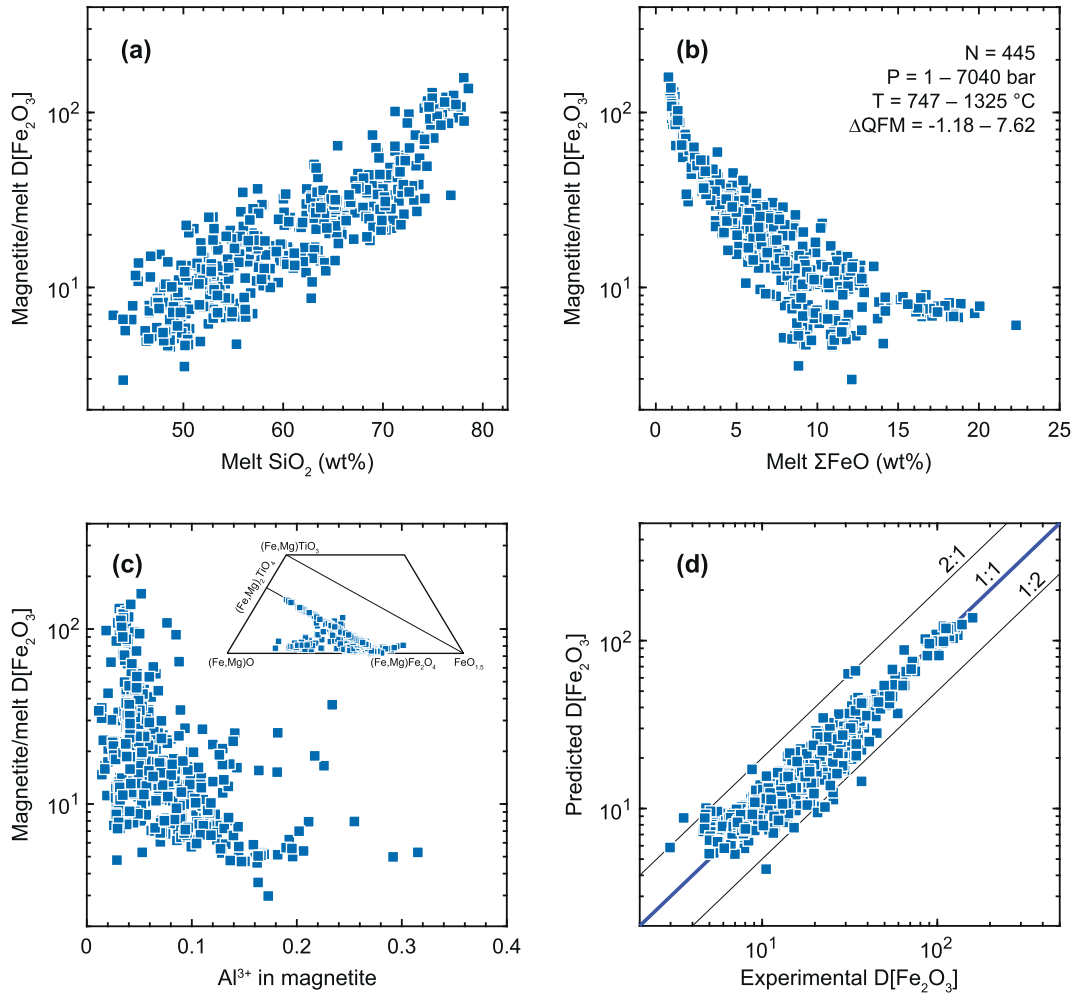

Fig. A2. Partition coefficients of Fe_2O_3 ($D[Fe_2O_3]$) between magnetite and melt from our compiled experiments (N = 445). Plots (a-b) show that SiO_2 and ΣFeO of the melt significantly affect $D[Fe_2O_3]$. Plot (c) demonstrates that $D[Fe_2O_3]$ also decreases with the increase of Al in magnetite. Plot (d) displays the comparison between the model predicted D values and the experimentally measured data. Inset in (c) shows the magnetite composition in a tetrahedral diagram. Details of the experiments are listed in Table S2.

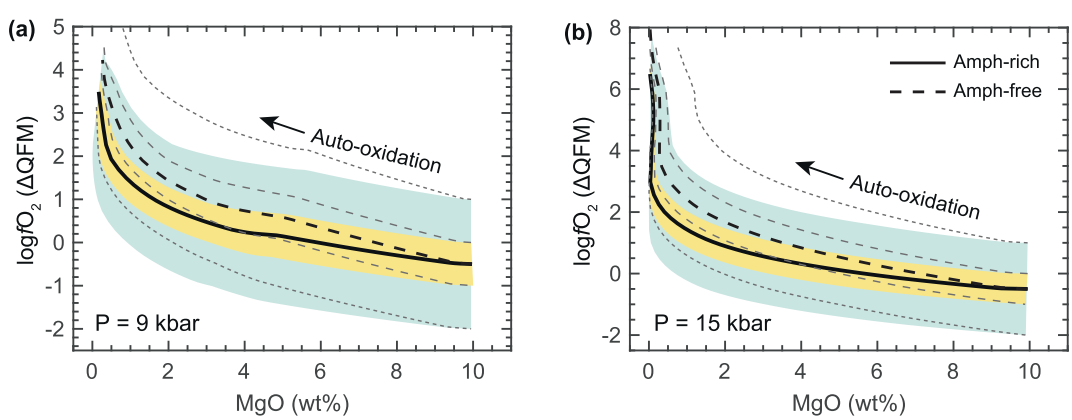

Fig. A3. Modeling results of melt redox variations during crystallization of amphibole-rich cumulate assemblages at 9 kbar (**a**) and 15 kbar (**b**). The initial melt compositions and fO_2 are the same as those in Fig. 4 under high-P conditions. Solid curves with color shades indicate the melt fO_2 variations due to crystallization of amphibole-rich cumulate assemblages. For comparison, dashed curves show the results in Fig. 4i and l from crystallization of amphibole-free cumulate assemblages.

Pichavant (2005; SP05), and Gaillard and and Scaillet (2014; GS14). These models have been proven to be remarkably useful for understanding volcanic degassing processes; however, they are not suitable for this study on crystallization and concurrent degassing of dynamically evolving magmatic systems.

SolEx incorporated three C-H-O-S species (i.e., CO_2, H_2O, SO_2) and assumed no redox interactions between the melt and gas. Using experimental data at 1150 °C and ΔQFM = ~+2.5 to +3, SolEx empirically calibrated the gas/melt distribution of S as a polynomial expression of P. Hence, SolEx is only applicable for oxidized

conditions. M03 and SP05 considered four C-H-O-S species (i.e., CO_2, H_2O, SO_2, and H_2S) but also ignored the gas–melt redox interactions. To constrain the total S content dissolved in the melt, SP05 calibrated an empirical S capacity model without considering different S species (S^{2-} vs S^{6+}) in the melt. The different S species, however, were resolved in M03 through two capacity models for sulfide and sulfate, respectively. In addition, the degassing of CO_2 and H_2O was constrained by SolEx and SP05 through the solubility models of Dixon et al. (1995) but was determined by M03 using the solubility models of Papale (1999). As demonstrated by Iacono-Marziano et al. (2012), Dixon et al.'s model underestimates the experimental CO_2 solubility data by ~400–2000 ppm at 1–4 kbar, while Papale's model overestimates the experimental H_2O solubility data by ~1–2 wt% at 1–2 kbar.

The GS14 and D-Compress models considered a more comprehensive list of C-H-O-S volatile species (i.e., H_2, H_2O, CO, CO_2, CH_4, H_2S, SO_2, and S_2) and incorporated the redox interaction between the melt and gas. In the GS14 model, the dissolved CO_2 and H_2O contents in the melt were estimated using the solubility models of Iacono-Marziano et al. (2012), which were also used in this study (see Section 4.1). As GS14 mainly focused on relatively reduced conditions, it omitted the dissolution of oxidized S species (S^{6+}) in the melt and therefore is not applicable to oxidized conditions. To constrain S^{2-} degassing, GS14 used the sulfide capacity of O'Neill and Mavrogenes (2002) that was experimentally calibrated at 1400 °C and 1 bar. D-Compress did not consider the different S species (S^{2-} vs S^{6+}) in the melt either. Instead of using sulfide/sulfate capacity models, D-Compress treated H_2S and SO_2 as soluble S species and then used a power law relation to empirically fit the solubility of H_2S or SO_2 as a function of its fugacity in the gas. This approach was also used by D-Compress to constrain the solubility of H_2O and CO_2. Because this approach did not incorporate the compositional effect, D-Compress implemented three sets of power law equations for rhyolites, basalts, and a phonolite melt (SiO_2 = 60 wt%), respectively. The calibration dataset for rhyolites has 72–78 wt% SiO_2, and that for basalts covers 47–49 wt% SiO_2. It is interesting to note that the three different sets of power law equations for volatile solubility also demonstrate a strong effect of melt composition on volatile degassing. Given the large compositional variation during magma differentiation (cf. Table S3), D-Compress is not suitable to model degassing driven by magma crystallization.

Another important caveat of these previous models (i.e., SolEx, M03, SP05, D-Compress, and GS14) is the lack of constraints on the saturation of S-rich phases, such as sulfide and sulfate, in the melt. Upon saturation, these S-rich phases can buffer the total S budget of the melt, impact the degassing of different S species, and further change the redox interactions between the melt and gas.

Appendix B. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.gca.2022.09.044.

References

Aiuppa, A., Inguaggiato, S., McGonigle, A.J.S., O'Dwyer, M., Oppenheimer, C., Padgett, M.J., Rouwet, D., Valenza, M., 2005. H_2S fluxes from Mt. Etna, Stromboli, and Vulcano (Italy) and implications for the sulfur budget at volcanoes. Geochim. Cosmochim. Acta 69, 1861–1871.

Ardia, P., Hirschmann, M.M., Withers, A.C., Stanley, B.D., 2013. Solubility of CH_4 in a synthetic basaltic melt, with applications to atmosphere-magma ocean-core partitioning of volatiles and to the evolution of the Martian atmosphere. Geochim. Cosmochim. Acta 114, 52–71.

Arevalo, R., McDonough, W.F., 2010. Chemical variations and regional diversity observed in MORB. Chem. Geol. 271, 70–85.

Armstrong, L.S., Hirschmann, M.M., Stanley, B.D., Falksen, E.G., Jacobsen, S.D., 2015. Speciation and solubility of reduced C-O-H-N volatiles in mafic melt: Implications for volcanism, atmospheric evolution, and deep volatile cycles in the terrestrial planets. Geochim. Cosmochim. Acta 171, 283–302.

Asimow, P.D., 2021. The petrological consequences of the estimated oxidation state of primitive MORB glass. Magma Redox Geochem., 139–154

Aulbach, S., Stagno, V., 2016. Evidence for a reducing Archean ambient mantle and its effects on the carbon cycle. Geology 44, 751–754.

Berry, A.J., Stewart, G.A., O'Neill, H.S.C., Mallmann, G., Mosselmans, J.F.W., 2018. A re-assessment of the oxidation state of iron in MORB glasses. Earth Planet. Sci. Lett. 483, 114–123.

Bézos, A., Guivel, C., La, C., Fougeroux, T., Humler, E., 2021. Unraveling the confusion over the iron oxidation state in MORB glasses. Geochim. Cosmochim. Acta 293, 28–39.

Bézos, A., Humler, E., 2005. The $Fe^{3+}/\Sigma Fe$ ratios of MORB glasses and their implications for mantle melting. Geochim. Cosmochim. Acta 69, 711–725.

Birner, S.K., Cottrell, E., Warren, J.M., Kelley, K.A., Davis, F.A., 2018. Peridotites and basalts reveal broad congruence between two independent records of mantle fO_2 despite local redox heterogeneity. Earth Planet. Sci. Lett. 494, 172–189.

Borisov, A., Behrens, H., Holtz, F., 2018. Ferric/ferrous ratio in silicate melts: a new model for 1 atm data with special emphasis on the effects of melt composition. Contrib. Mineral. Petrol. 173, 1–15.

Brounce, M., Stolper, E., Eiler, J., 2017. Redox variations in Mauna Kea lavas, the oxygen fugacity of the Hawaiian plume, and the role of volcanic gases in Earth's oxygenation. Proc. Natl. Acad. Sci. U. S. A. 114, 8997–9002.

Burgisser, A., Scaillet, B., Harshvardhan, 2008. Chemical patterns of erupting silicic magmas and their influence on the amount of degassing during ascent. J. Geophys. Res. Solid Earth 113.

Burgisser, A., Scaillet, B., 2007. Redox evolution of a degassing magma rising to the surface. Nature 445, 194–197.

Byers, C.D., Muenow, D.W., Garcia, M.O., 1983. Volatiles in basalts and andesites from the Galapagos Spreading Center, 85° to 86°W. Geochim. Cosmochim. Acta 47, 1551–1558.

Byers, C.D., Christie, D.M., Muenow, D.W., Sinton, J.M., 1984. Volatile contents and ferric-ferrous ratios of basalt, ferrobasalt, andesite and rhyodacite glasses from the Galapagos 95.5°W propagating rift. Geochim. Cosmochim. Acta 48, 2239–2245.

Canfield, D.E., 2005. The early history of atmospheric oxygen: Homage to Robert M. Garrels. Annu. Rev. Earth Planet. Sci. https://doi.org/10.1146/annurev.earth.33.092203.122711.

Canil, D., 1997. Vanadium partitioning and the oxidation state of Archaean komatiite magmas. Nature 389, 842–845.

Carmichael, I.S.E., 1991. The redox states of basic and silicic magmas: a reflection of their source regions? Contrib. Mineral. Petrol. 106, 129–141.

Chase, M.W., 1996. NIST-JANAF thermochemical tables for the iodine oxides. J. Phys. Chem. Ref. Data 25, 1229–1339.

Christie, D.M., Carmichael, I.S.E., Langmuir, C.H., 1986. Oxidation states of mid-ocean ridge basalt glasses. Earth Planet. Sci. Lett. 79, 397–411.

Cicconi, M.R., Le Losq, C., Moretti, R., Neuville, D.R., 2020. Magmas are the largest repositories and carriers of earth's redox processes. Elements 16, 173–178.

Cottrell, E., Birner, S.K., Brounce, M., Davis, F.A., Waters, L.E., Kelley, K.A., 2021. Oxygen fugacity across tectonic settings. Magma Redox Geochem., 33–61

Cottrell, E., Kelley, K.A., 2011. The oxidation state of Fe in MORB glasses and the oxygen fugacity of the upper mantle. Earth Planet. Sci. Lett. 305, 270–282.

Davis, F.A., Cottrell, E., 2018. Experimental investigation of basalt and peridotite oxybarometers: Implications for spinel thermodynamic models and Fe3+ compatibility during generation of upper mantle melts. Am. Mineral. 103, 1056–1067.

Delano, J.W., 2001. Redox history of the Earth's interior since sim;3900 Ma: Implications for prebiotic molecules. Orig. Life Evol. Biosph. 31, 311–341.

Dhuime, B., Wuestefeld, A., Hawkesworth, C.J., 2015. Emergence of modern continental crust about 3 billion years ago. Nat. Geosci. 8, 552–555.

Ding, S., Dasgupta, R., 2017. The fate of sulfide during decompression melting of peridotite–implications for sulfur inventory of the MORB-source depleted upper mantle. Earth Planet. Sci. Lett. 459, 183–195.

Dixon, J.E., Stolper, E.M., Holloway, J.R., 1995. An experimental study of water and carbon dioxide solubilities in mid-ocean ridge basaltic liquids. Part I: Calibration and solubility models. J. Petrol. 36, 1607–1631.

Durrheim, R.J., Mooney, W.D., 1991. Archean and Proterozoic crustal evolution: evidence from crustal seismology. Geology 19, 606–609.

Eguchi, J., Seales, J., Dasgupta, R., 2020. Great Oxidation and Lomagundi events linked by deep cycling and enhanced degassing of carbon. Nat. Geosci. 13, 71–76.

Farquhar, J., Wing, B.A., 2003. Multiple sulfur isotopes and the evolution of the atmosphere. Earth Planet. Sci. Lett. 213, 1–13.

Fortin, M.-A., Riddle, J., Desjardins-Langlais, Y., Baker, D.R., 2015. The effect of water on the sulfur concentration at sulfide saturation (SCSS) in natural melts. Geochim. Cosmochim. Acta 160, 100–116.

Francis, R.D., 1990. Sulfide globules in mid-ocean ridge basalts (MORB), and the effect of oxygen abundance in FeSO liquids on the ability of those liquids to partition metals from MORB and komatiite magmas. Chem. Geol. 85, 199–213.

Gaillard, F., Scaillet, B., 2014. A theoretical framework for volcanic degassing chemistry in a comparative planetology perspective and implications for planetary atmospheres. Earth Planet. Sci. Lett. 403, 307–316.
Gaillard, F., Schmidt, B., Mackwell, S., McCammon, C., 2003. Rate of hydrogen-iron redox exchange in silicate melts and glasses. Geochim. Cosmochim. Acta 67, 2427–2441.
Gaillard, F., Scaillet, B., Arndt, N.T., 2011. Atmospheric oxygenation caused by a change in volcanic degassing pressure. Nature 478, 229–232.
Ghiorso, M.S., Hirschmann, M.M., Reiners, P.W., Kress, V.C., 2002. The pMELTS: A revision of MELTS for improved calculation of phase relations and major element partitioning related to partial melting of the mantle to 3 GPa. Geochem. Geophys. Geosyst. 3, 1–35.
Holland, H.D., 2002. Volcanic gases, black smokers, and the great oxidation event. Geochim. Cosmochim. Acta 66, 3811–3826.
Holland, H.D., 2009. Why the atmosphere became oxygenated: A proposal. Geochim. Cosmochim. Acta 73, 5241–5255.
Holloway, J.R., 1987. Igneous fluids. Rev. Mineral. Geochem. 17, 211–233.
Holloway, J.R., Blank, J.G., 1994. Application of experimental results to C-O-H species in natural melts. Rev. Mineral. 30, 187–230.
Holloway, J.R., Pan, V., Gudmundsson, G., 1992. High-pressure fluid-absent melting experiments in the presence of graphite: oxygen fugacity, ferric/ferrous ratio and dissolved CO_2. Eur. J. Mineral. 4, 105–114.
Iacono-Marziano, G., Morizet, Y., Le Trong, E., Gaillard, F., 2012. New experimental data and semi-empirical parameterization of H_2O-CO_2 solubility in mafic melts. Geochim. Cosmochim. Acta 97, 1–23.
Iacovino, K., 2015. Linking subsurface to surface degassing at active volcanoes: A thermodynamic model with applications to Erebus volcano. Earth Planet. Sci. Lett. 431, 59–74.
Jayasuriya, K.D., O'Neill, H.S.C., Berry, A.J., Campbell, S.J., 2004. A Mössbauer study of the oxidation state of Fe in silicate melts. Am. Mineral. 89, 1597–1609.
Jenner, F.E., O'Neill, H.S.C., Arculus, R.J., Mavrogenes, J.A., 2012. The magnetite crisis in the evolution of arc-related magmas and the initial concentration of Au, Ag and Cu. J. Petrol. 53, 1089.
Kadoya, S., Catling, D.C., Nicklas, R.W., Puchtel, I.S., Anbar, A.D., 2020a. Mantle data imply a decline of oxidizable volcanic gases could have triggered the Great Oxidation. Nat. Commun. 11.
Kadoya, S., Catling, D.C., Nicklas, R.W., Puchtel, I.S., Anbar, A.D., 2020b. Mantle cooling causes more reducing volcanic gases and gradual reduction of the atmosphere. Geochemical Perspect. Lett. 13, 25–29.
Kasting, J.F., 2013. What caused the rise of atmospheric O_2? Chem. Geol. https://doi.org/10.1016/j.chemgeo.2013.05.039.
Kasting, J.F., Eggler, D.H., Raeburn, S.P., 1993. Mantle redox evolution and the oxidation state of the Archean atmosphere. J. Geol. 101, 245–257.
Keller, C.B., Schoene, B., 2012. Statistical geochemistry reveals disruption in secular lithospheric evolution about 2.5Gyr ago. Nature 485, 490–493.
Kelley, K.A., Cottrell, E., 2012. The influence of magmatic differentiation on the oxidation state of Fe in a basaltic arc magma. Earth Planet. Sci. Lett. 329–330, 109–121.
Kilinc, A., Carmichael, I.S.E., Rivers, M.L., Sack, R.O., 1983. The ferric-ferrous ratio of natural silicate liquids equilibrated in air. Contrib. Mineral. Petrol. 83, 136–140.
King, P.L., Hervig, R.L., Holloway, J.R., Delaney, J.S., Dyar, M.D., 2000. Partitioning of Fe^{3+}/Fe(total) between amphibole and basanitic melt as a function of oxygen fugacity. Earth Planet. Sci. Lett. 178, 97–112.
Kress, V.C., Carmichael, I.S.E., 1991. The compressibility of silicate liquids containing Fe_2O_3 and the effect of composition, temperature, oxygen fugacity and pressure on their redox states. Contrib. Mineral. Petrol. 108, 82–92.
Kump, L.R., Barley, M.E., 2007. Increased subaerial volcanism and the rise of atmospheric oxygen 2.5 billion years ago. Nature 448, 1033–1036.
Lee, C.-T.-A., Brandon, A.D., Norman, M., 2003. Vanadium in peridotites as a proxy for paleo-fO_2 during partial melting. Geochim. Cosmochim. Acta 67, 3045–3064.
Lee, C.-T.-A., Luffi, P., Le Roux, V., Dasgupta, R., Albaréde, F., Leeman, W.P., 2010. The redox state of arc mantle using Zn/Fe systematics. Nature 468, 681–685.
Lee, C.-T.-A., Yeung, L.Y., McKenzie, N.R., Yokoyama, Y., Ozaki, K., Lenardic, A., 2016. Two-step rise of atmospheric oxygen linked to the growth of continents. Nat. Geosci. 9, 417–424.
Lesne, P., Kohn, S.C., Blundy, J., Witham, F., Botcharnikov, R.E., Behrens, H., 2011. Experimental simulation of closed-system degassing in the system basalt-H_2O-CO_2-S-Cl. J. Petrol. 52, 1737–1762.
Li, Z.X., Lee, C.-T.-A., 2004. The constancy of upper mantle fO_2 through time inferred from V/Sc ratios in basalts. Earth Planet. Sci. Lett. 228, 483–493.
Losq, C.L., Berry, A.J., Kendrick, M.A., Neuville, D.R., O'Neill, H.S.C., 2019. Determination of the oxidation state of iron in Mid-Ocean Ridge basalt glasses by Raman spectroscopy. Am. Mineral. 104, 1032–1042.
Lyons, T.W., Reinhard, C.T., Planavsky, N.J., 2014. The rise of oxygen in Earth's early ocean and atmosphere. Nature. https://doi.org/10.1038/nature13068.
Mallmann, G., O'Neill, H.S.C., 2009. The crystal/melt partitioning of V during mantle melting as a function of oxygen fugacity compared with some other elements (Al, P, Ca, Sc, Ti, Cr, Fe, Ga, Y, Zr and Nb). J. Petrol. 50, 1765–1794.
Mangiacapra, A., Moretti, R., Rutherford, M., Civetta, L., Orsi, G., Papale, P., 2008. The deep magmatic system of the Campi Flegrei caldera (Italy). Geophys. Res. Lett. 35.
Martel, C., Pichavant, M., Holtz, F., Scaillet, B., Bourdier, J.-L., Traineau, H., 1999. Effects of f O_2 and H_2O on andesite phase relations between 2 and 4 kbar. J. Geophys. Res. Solid Earth 104, 29453–29470.
Mathez, E.A., 1984. Influence of degassing on oxidation states of basaltic magmas. Nature 310, 371–375.
Mooney, W.D., Laske, G., Masters, T.G., 1998. CRUST 5.1: A global crustal model at 5° × 5°. J. Geophys. Res. Solid Earth 103, 727–747.
Moré, J.J., Garbow, B.S., Hillstrom, K.E., 1980. User guide for MINPACK-1.
Moretti, R., 2005. Polymerisation, basicity, oxidation state and their role in ionic modelling of silicate melts. Ann. Geophys. 48, 583–608.
Moretti, R., Ottonello, G., 2003. Polymerization and disproportionation of iron and sulfur in silicate melts: Insights from an optical basicity-based approach. J. Non. Cryst. Solids 323, 111–119.
Moretti, R., Papale, P., 2004. On the oxidation state and volatile behavior in multicomponent gas-melt equilibria. Chem. Geol. 213, 265–280.
Moretti, R., Papale, P., Ottonello, G., 2003. A model for the saturation of C-O-H-S fluids in silicate melts. Geol. Soc. Spec. Publ. 213, 81–101.
Moussallam, Y., Oppenheimer, C., Scaillet, B., Gaillard, F., Kyle, P., Peters, N., Hartley, M., Berlo, K., Donovan, A., 2014. Tracking the changing oxidation state of Erebus magmas, from mantle to surface, driven by magma ascent and degassing. Earth Planet. Sci. Lett. 393, 200–209.
Moussallam, Y., Edmonds, M., Scaillet, B., Peters, N., Gennaro, E., Sides, I., Oppenheimer, C., 2016. The impact of degassing on the oxidation state of basaltic magmas: A case study of Kīlauea volcano. Earth Planet. Sci. Lett. 450, 317–325.
Moussallam, Y., Oppenheimer, C., Scaillet, B., 2019. On the relationship between oxidation state and temperature of volcanic gas emissions. Earth Planet. Sci. Lett. 520, 260–267.
Myers, J., Eugster, H.P., 1983. The system Fe-Si-O: Oxygen buffer calibrations to 1,500K. Contrib. Mineral. Petrol. 82, 75–90.
Nash, W.M., Smythe, D.J., Wood, B.J., 2019. Compositional and temperature effects on sulfur speciation and solubility in silicate melts. Earth Planet. Sci. Lett. 507, 187–198.
Nicklas, R.W., Puchtel, I.S., Ash, R.D., Piccoli, P.M., Hanski, E., Nisbet, E.G., Waterton, P., Pearson, D.G., Anbar, A.D., 2019. Secular mantle oxidation across the Archean-Proterozoic boundary: Evidence from V partitioning in komatiites and picrites. Geochim. Cosmochim. Acta 250, 49–75.
Nzotta, M.M., Sichen, D., Seetharaman, S., 1999. Study of the sulfide capacities of iron-oxide containing slags. Metall. Mater. Trans. B Process Metall. Mater. Process. Sci. 30, 909–920.
O'Neill, H.S.C., Berry, A.J., Mallmann, G., 2018. The oxidation state of iron in Mid-Ocean Ridge Basaltic (MORB) glasses: Implications for their petrogenesis and oxygen fugacities. Earth Planet. Sci. Lett. 504, 152–162.
O'Neill, H.S.C., Mavrogenes, J.A., 2002. The sulfide capacity and the sulfur content at sulfide saturation of silicate melts at 1400°C and 1 bar. J. Petrol. 43, 1049–1087.
Oppenheimer, C., Moretti, R., Kyle, P.R., Eschenbacher, A., Lowenstern, J.B., Hervig, R. L., Dunbar, N.W., 2011. Mantle to surface degassing of alkalic magmas at Erebus volcano, Antarctica. Earth Planet. Sci. Lett. 306, 261–271.
Papale, P., 1999. Modeling of the solubility of a two-component H_2O + CO_2 fluid in silicate liquids. Am. Mineral. 84, 477–492.
Patten, C., Barnes, S.J., Mathez, E.A., 2012. Textural variations in morb sulfide droplets due to differences in crystallization history. Can. Mineral. 50, 675–692.
Peng, D.Y., Robinson, D.B., 1976. A new two-constant equation of state. Ind. Eng. Chem. Fundam. 15, 59–64.
Planavsky, N.J., Asael, D., Hofmann, A., Reinhard, C.T., Lalonde, S.V., Knudsen, A., Wang, X., Ossa Ossa, F., Pecoits, E., Smith, A.J.B., Beukes, N.J., Bekker, A., Johnson, T.M., Konhauser, K.O., Lyons, T.W., Rouxel, O.J., 2014. Evidence for oxygenic photosynthesis half a billion years before the Great Oxidation Event. Nat. Geosci. 7, 283–286.
Plank, T., Manning, C.E., 2019. Subducting carbon. Nature 574, 343–352.
Rudra, A., Cottrell, E., Hirschmann, M.M., 2021. Experimental determination of ferric iron partitioning between pyroxene and melt at 100 kPa. Chem. Geol. 584, 120532.
Rudra, A., Hirschmann, M.M., 2022. Fe^{3+} partitioning between clinopyroxene and silicate melt at 1–2.5 GPa: Implications for Fe^{3+} content of MORB and OIB source mantle. Geochim. Cosmochim. Acta 328, 258–279.
Scaillet, B., Evans, B.W., 1999. The 15 June 1991 eruption of Mount Pinatubo. I. Phase equilibria and pre-eruption P-T-fO_2-fH_2O conditions of the dacite magma. J. Petrol. 40, 381–411.
Scaillet, B., Pichavant, M., 2005. A model of sulphur solubility for hydrous mafic melts: Application to the determination of magmatic fluid compositions of Italian volcanoes. Ann. Geophys. 48, 671–698.
Shi, P., Saxena, S.K., 1992. Thermodynamic modeling of the C-H-O-S fluid system. Am. Mineral. 77, 1038–1049.
Sisson, T.W., Ratajeski, K., Hankins, W.B., Glazner, A.F., 2005. Voluminous granitic magmas from common basaltic sources. Contrib. Mineral. Petrol. 148, 635–661.
Smith, P.M., Asimow, P.D., 2005. Adiabat-1ph: A new public front-end to the MELTS, pMELTS, and pHMELTS models. Geochem. Geophys. Geosyst. 6.
Smythe, D.J., Wood, B.J., Kiseeva, E.S., 2017. The S content of silicate melts at sulfide saturation: new experiments and a model incorporating the effects of sulfide composition. Am. Mineral. 102, 795–803.
Sun, C., Dasgupta, R., 2020. Thermobarometry of CO2-rich, silica-undersaturated melts constrains cratonic lithosphere thinning through time in areas of kimberlitic magmatism. Earth Planet. Sci. Lett. 550, 116549.

Szwillus, W., Afonso, J.C., Ebbing, J., Mooney, W.D., 2019. Global crustal thickness and velocity structure from geostatistical analysis of seismic data. J. Geophys. Res. Solid Earth 124, 1626–1652.
Tang, M., Erdman, M., Eldridge, G., Lee, C.T.A., 2018. The redox "filter" beneath magmatic orogens and the formation of continental crust. Sci. Adv. 4.
Tang, M., Lee, C.T.A., Rudnick, R.L., Condie, K.C., 2020. Rapid mantle convection drove massive crustal thickening in the late Archean. Geochim. Cosmochim. Acta 278, 6–15.
Wallace, P., Carmichael, I.S.E., 1992. Sulfur in basaltic magmas. Geochim. Cosmochim. Acta 56, 1863–1874.
Witham, F., Blundy, J., Kohn, S.C., Lesne, P., Dixon, J., Churakov, S.V., Botcharnikov, R., 2012. SolEx: A model for mixed COHSCl-volatile solubilities and exsolved gas compositions in basalt. Comput. Geosci. 45, 87–97.
Zajacz, Z., Tsay, A., 2019. An accurate model to predict sulfur concentration at anhydrite saturation in silicate melts. Geochim. Cosmochim. Acta 261, 288–304.
Zhang, H.L., Cottrell, E., Solheid, P.A., Kelley, K.A., Hirschmann, M.M., 2018. Determination of $Fe^{3+}/\Sigma Fe$ of XANES basaltic glass standards by Mössbauer spectroscopy and its application to the oxidation state of iron in MORB. Chem. Geol. 479, 166–175.